PHEXARA LIMITED | UNITED KINGDOM

Vet2Employ

Simulated Working Environment

Week One Work Brief for four professional tracks: Data Analyst, Software Developer and Engineer, Cybersecurity Analyst and Data Engineer

1 **Data Analyst**
Kestrel Lane Home and Garden Ltd

2 **Software Developer and Engineer**
Northwold Couriers Ltd

3 **Cybersecurity**
Ashbourne Valley Credit Union

4 **Data Engineering**
Voltaway Charging Networks Ltd

Document details
Document type: Programme Work Brief
Cohort: Vet2Employ Simulation, Week One
Version 1.0, issued October 2026
Classification: Programme Confidential

At a glance
Every track is a full working week: plan for 20 to 28 focused hours and log every one of them.

Every client, person, dataset and incident in this brief is fictitious and created for training purposes by Phexara Limited.

Document control

Item	Detail
Document title	Vet2Employ Simulated Working Environment: Week One Work Brief
Issuing organisation	Phexara Limited, United Kingdom
Programme	Vet2Employ (Vetted to Employ) Simulated Working Environment
Version	1.0
Issue date	12 October 2026
Audience	Programme participants, track supervisors, peer reviewers and assessors
Tracks covered	Data Analyst; Software Developer and Engineer; Cybersecurity; Data Engineering
Minimum effort	20 logged hours per participant across one working week
Classification	Programme Confidential. Do not publish the brief itself; you may publish your own work in a portfolio once the week has been assessed.
Review cycle	Reviewed after every cohort by the Vet2Employ Programme Office

How to read this brief

This brief has two parts. **Part A** (Sections 1 to 6) applies to every participant regardless of track. It explains how the simulated working environment operates, how your week is structured, the standards you must meet, how to submit your work and how you will be assessed. **Part B** (Sections 7 to 10) contains the four track briefs. You only need to complete the brief for the track you have been assigned, although you are encouraged to read the others so that you understand what your colleagues in other disciplines are working on.

Every track brief follows the same pattern so that you always know where to look:

1. **A welcome note** from your track supervisor that sets the tone for the week.
2. **A detailed overview of the client and the problem** you are being asked to solve, including the business context, the pain points, the stakeholders and the constraints.
3. **Your working materials**, which describe the data, systems or evidence that the client has provided.
4. **A week at a glance** table showing how the effort is distributed from Monday to Friday.
5. **A list of sequential activities.** Each activity has an objective, ordered steps and clear completion criteria. Activities build on one another, so complete them in the order given.
6. **The required deliverables**, with file names, formats, due points and acceptance criteria.
7. **An assessment rubric** that shows exactly what good looks like.

Diagrams in this brief

Diagrams use annotated text boxes. Shaded call out boxes with a pointer highlight something you should pay attention to, such as a known weakness, a design decision or a common mistake. Treat each call out as a hint from your supervisor.

Contents

This brief is in two parts. Part A applies to every participant. Part B contains the four track briefs; complete only the one for your assigned track.

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Every client, person, dataset and scenario in this brief is fictitious and created for training purposes by Phexara Limited.

1. Welcome to Vet2Employ

Dear Participant,

On behalf of everyone at Phexara Limited, welcome to Vet2Employ. We are genuinely pleased that you have chosen to spend the coming week with us, and we want you to know from the very first page that this is not a course, a quiz or a classroom exercise. It is a working week.

We built Vet2Employ because we kept meeting talented people who had completed courses, earned certificates and built tutorial projects, yet still struggled to show an employer that they could walk into a team on a Monday morning and deliver something useful by Friday afternoon. Employers do not only hire for knowledge. They hire for judgement, for communication, for the ability to work with incomplete information and for the discipline to finish what you start. Those qualities are difficult to demonstrate on a CV, and they are almost impossible to learn without practice. Vet2Employ exists to give you that practice in a safe environment where mistakes are expected and feedback is generous.

During this week you will join a simulated Phexara delivery team that has been engaged by a fictitious client. The client has a real business problem, imperfect data or systems, busy stakeholders and a deadline. Your supervisor will behave as a team lead would in a consultancy or an internal technology department. You will attend short check ins, raise questions, make decisions, record your reasoning and present your results. You will be treated as a junior professional, and we will expect you to behave like one.

Please read Part A of this brief carefully before you open your track brief. It contains the rules of the environment, the way you will be assessed and the standards that apply to everyone. Then read your track brief from start to finish before you begin any work. Experienced professionals always read the whole brief first, because the deliverables at the end shape the decisions you make at the beginning.

Finally, remember that asking a good question is a professional skill, not a weakness. If something in the brief is unclear, ask. If the data or the system does not behave the way you expect, record what you found and tell us. Some of those surprises have been placed there on purpose.

We look forward to seeing what you build. Have a productive and rewarding week.

The Vet2Employ Programme Office
Phexara Limited, United Kingdom

2. About the Simulated Working Environment

2.1 Purpose of the programme

Vet2Employ is a structured, assessed simulation of professional work in four technology disciplines. Each cohort completes a one week engagement for a fictitious client. The engagement is designed to reproduce the conditions of a real workplace as closely as is practical: a written brief, an accountable supervisor, a fixed timeline, messy inputs, stakeholders with competing priorities and a requirement to communicate results to people who are not specialists.

At the end of the week each participant receives a written assessment against a common competency framework, detailed feedback on every deliverable and, where earned, a Vet2Employ Work Readiness Statement that can be shared with prospective employers. The work you produce becomes a portfolio piece that you can discuss confidently in interviews because you will have done it under realistic conditions.

2.2 Programme objectives

By the end of the week you should be able to demonstrate that you can:

- interpret a business brief and translate it into a structured, time boxed plan of work;
- apply the core technical skills of your discipline to an unfamiliar problem;
- identify, document and handle imperfect inputs rather than ignoring them;
- make and justify decisions, recording assumptions where information is missing;
- produce deliverables to a professional standard of quality, naming and presentation;
- communicate findings clearly to both technical and nontechnical audiences;
- manage your own time, track your effort honestly and meet a deadline.

2.3 Who is who in the simulation

You will interact with several roles during the week. Some are played by Phexara staff and some are fictitious personas whose views are described in the brief. The diagram below shows how they relate to you.

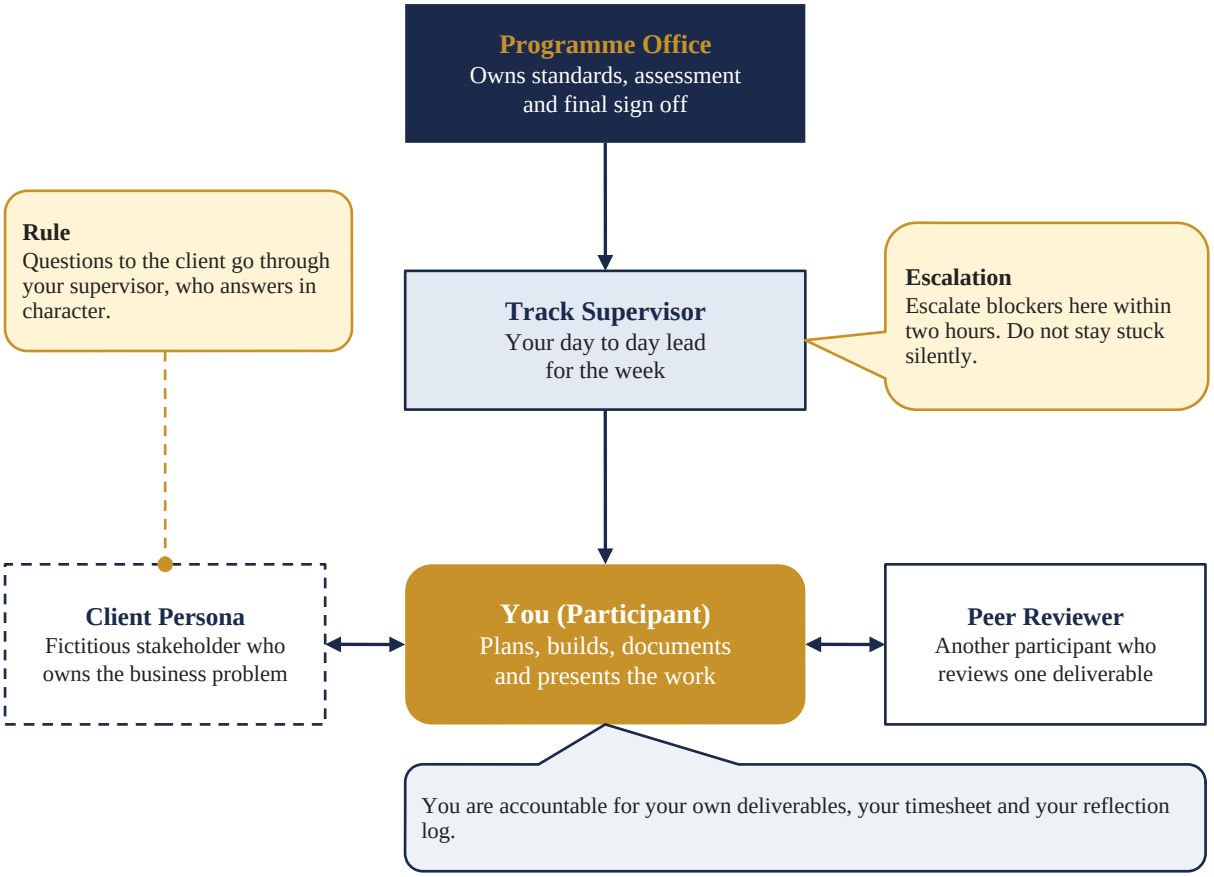


Figure 2.1. Roles and reporting lines inside the Vet2Employ simulation

Role	What they do	How you work with them
Programme Office	Sets the rules of the simulation, moderates assessment and issues the Work Readiness Statement.	You will rarely contact them directly. Raise concerns about fairness or wellbeing with them at any time.
Track Supervisor	A Phexara practitioner in your discipline who leads your team for the week, runs check ins and marks your deliverables.	Attend every check in, send your daily update and escalate blockers promptly.
Client Persona	A fictitious manager at the client organisation. Their priorities and opinions are described in your track brief.	Submit written clarification questions through your supervisor, who replies in character within half a working day.
Peer Reviewer	Another participant, usually from a different track, who reviews one of your deliverables on Thursday.	Exchange work respectfully, give specific feedback and record what you changed as a result.

2.4 What makes this a simulation and not a course

There are no lectures and no model answers handed out during the week. You will be given a brief, materials and a supervisor. How you approach the problem is up to you, provided that you meet the deliverable requirements and the professional standards in Section 4. Several deliberate complications have been placed in every track: inconsistent data, ambiguous requirements, a stakeholder who changes their mind or an artefact that contradicts another. Your response to these complications is assessed just as carefully as your technical output.

Professionals do not pretend that problems are not there; they notice them, record them, decide what to do and explain why.

3. How Your Week Runs

3.1 The weekly rhythm

Every track runs on the same weekly rhythm so that supervisors, peer reviewers and the Programme Office can coordinate. The rhythm is shown below. Fixed events are marked; the rest of the time is yours to plan, provided that you complete the activities in sequence and log at least twenty hours.

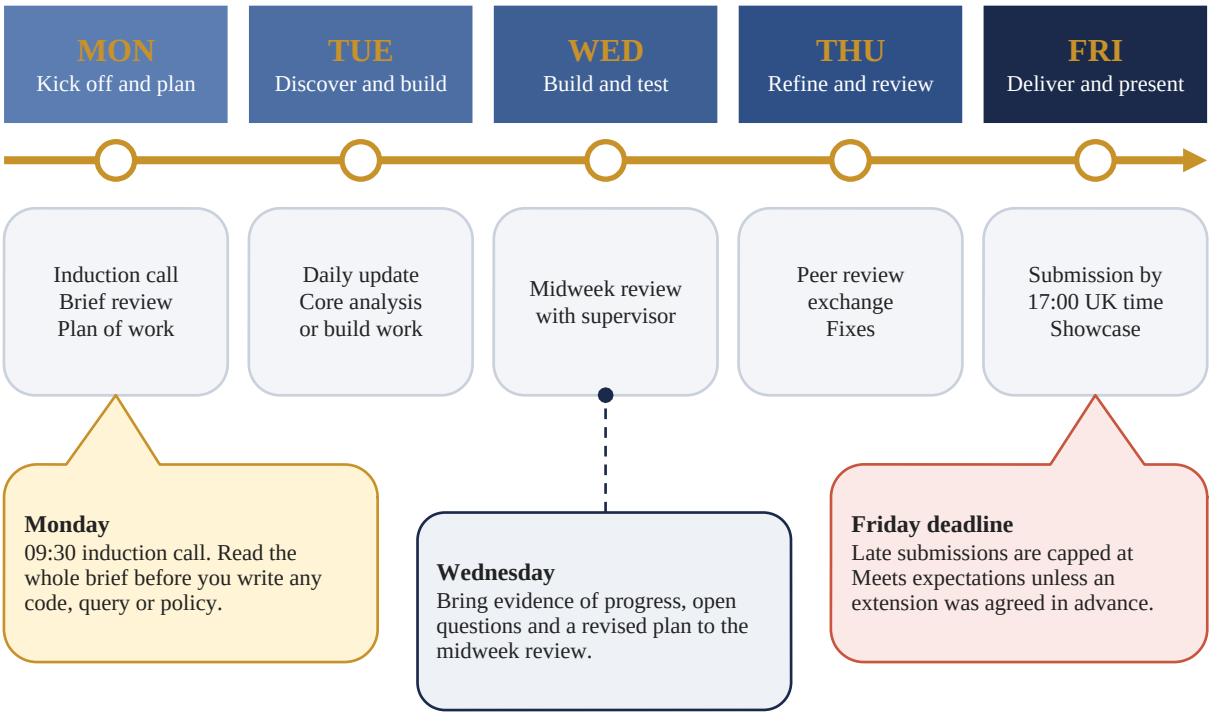


Figure 3.1. The Vet2Employ weekly rhythm with fixed checkpoints

3.2 Fixed events

Event	When	Duration	What happens
Induction call	Monday 09:30	45 minutes	Welcome, introductions, walkthrough of the brief, access to materials and confirmation of your track.
Daily written update	Tuesday to Friday by 10:00	10 minutes	A short written update using the template in Appendix B: done yesterday, planned today, blockers.
Client question window	Monday 14:00 to Tuesday 12:00	Open	Submit written clarification questions to the client persona through your supervisor. Late questions may not be answered.
Midweek review	Wednesday, booked slot	20 minutes	Show your progress, discuss risks and agree any change of scope with your supervisor.

Event	When	Duration	What happens
Peer review exchange	Thursday by 12:00	60 minutes	Swap one deliverable with your assigned peer, give written feedback and act on what you receive.
Final submission	Friday by 17:00 UK time	Not applicable	Upload every deliverable to your submission folder using the naming convention in Section 5.
Showcase	Friday 15:30 or the following Monday	10 minutes each	Present your results to the cohort and supervisors as if they were the client.

3.3 Effort and the timesheet

Every track is designed to be completed in **at least twenty hours** of focused work, and the planned effort for each track sits between twenty six and twenty eight hours. You must record your time in the timesheet provided in Appendix A. Record time against the activity number, in blocks of no less than fifteen minutes, and describe what you actually did. Your timesheet is part of the Professionalism assessment and is reviewed alongside your deliverables.

Honest timesheets matter more than impressive ones. If an activity takes far longer than planned, record it and explain why in your reflection log. Estimation is a skill that improves only when you compare estimates with reality. If you finish early, use the stretch goals that appear in each track rather than padding your hours.

3.4 Communication norms

- Use the cohort collaboration workspace allocated to you at induction (for example the Phexara ProWin workspace or the channel your supervisor names). Keep project discussion out of private messages so that decisions are visible.
- Write in plain, professional English. Lead with the point, then the detail.
- When you ask a question, state what you have already tried and what you think the answer might be.
- Acknowledge messages from your supervisor within two working hours during the core day of 10:00 to 16:00 UK time.
- If you are blocked for more than two hours, escalate. Being stuck silently is the most common reason participants miss the Friday deadline.
- Record significant decisions in your decision log (part of your reflection log) with the date, the decision, the options you considered and the reason for your choice.

4. Professional Standards and Rules of Work

4.1 Professional conduct

You are expected to behave as you would in a paid role at a reputable UK employer. That means punctuality for fixed events, respectful communication with everyone in the cohort, honesty about progress and ownership of your mistakes. Disrespectful behaviour, harassment or discrimination of any kind will result in removal from the cohort.

4.2 Confidentiality and data handling

All client materials in Vet2Employ are fictitious, but you must handle them as if they were real. This builds the habits that UK employers expect under the UK General Data Protection Regulation and the Data Protection Act 2018.

- Store materials only in the locations named at induction. Do not upload client datasets to public repositories, public file sharing services or public forums.
- Treat any field that looks like personal data (names, emails, postcodes, account numbers) as personal data. Minimise it, mask it in screenshots and never paste it into external tools.
- Delete local copies of the materials within thirty days of the end of the cohort unless the Programme Office confirms that a dataset has been released for portfolio use.
- You may publish your own code, dashboards and reports in a portfolio after assessment, provided that you state clearly that the client and data are fictitious and created by Phexara Limited.

4.3 Use of artificial intelligence tools

Phexara uses artificial intelligence tools in its own work and we expect our graduates to use them responsibly. You may use AI assistants to help you learn, to explain errors, to draft boilerplate and to review your writing. You may not present AI generated work as your own judgement. The following rules apply:

1. Declare every substantial use of an AI tool in your reflection log: which tool, for what purpose and how you checked the result.
2. You must be able to explain every line of code, every formula and every recommendation that you submit. Supervisors will ask you to do so during the showcase.
3. Never paste client data, credentials or log extracts that contain personal data into an external AI service. Use synthetic examples instead.
4. Where an AI tool produces a factual claim, verify it against an authoritative source before you rely on it.

4.4 Academic and professional integrity

Collaboration is encouraged; copying is not. You may discuss approaches with other participants, but every deliverable you submit must be your own work. Reuse of code from public sources is acceptable if it is licensed for reuse, clearly attributed in a comment or reference list and adapted to the problem. Submitting another participant's work, or work from a previous cohort, will result in a fail for the week.

4.5 Safe and legal practice

Some activities, particularly in the Cybersecurity track, involve tools that could cause harm if misused. You may only test systems that the brief explicitly places in scope. Testing any other system, including systems belonging to Phexara, your employer, your school or any third party, is prohibited and may be a criminal offence under the Computer Misuse Act 1990. If you are unsure whether something is in scope, stop and ask.

4.6 Wellbeing and reasonable adjustments

The week is demanding but it should not be harmful. Take regular breaks, avoid working late into the night and tell the Programme Office if you need a reasonable adjustment, such as additional time, an alternative presentation format or written rather than spoken check ins. Adjustments are confidential and do not affect your assessment.

5. Submission Standards

5.1 Naming convention

Every file you submit must follow the naming convention below. Consistent naming is how real teams find work quickly, and it is assessed under Professionalism.

> **Required pattern**
>
> **V2E_[Track]_[Surname]_[DeliverableRef]_[ShortName]_v[Number].[extension]**
>
> Example: V2E_DA_Okafor_D4_KPIDashboard_v2.pbix
>
> Track codes: DA (Data Analyst), SE (Software Developer and Engineer), CY (Cybersecurity), DE (Data Engineering). Use underscores, never spaces.

5.2 Standard submission folder

Your submission folder must use the structure below. Each track brief lists the specific files expected in each subfolder.

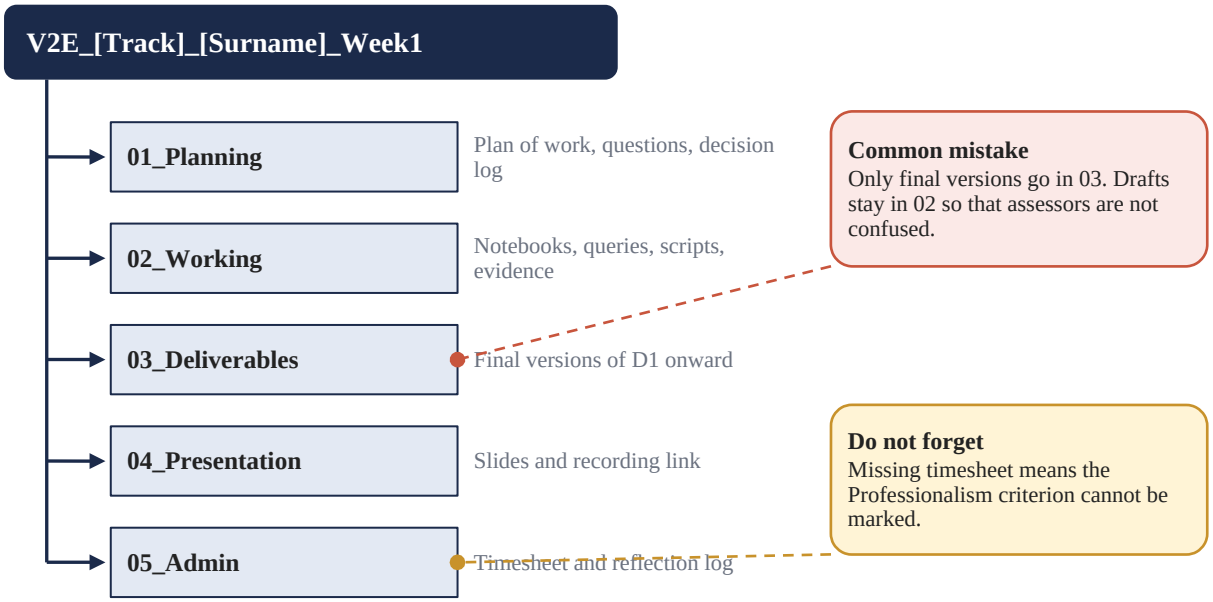


Figure 5.1. Standard submission folder structure with annotations

5.3 Format rules

Artefact type	Accepted formats	Rules
Written reports	PDF, plus the editable source (.docx)	Times New Roman or another readable serif or sans serif typeface, numbered pages, a title page and an executive summary.
Slides	PDF and the editable source (.pptx)	No more than the slide limit stated in your track. One message per slide. Readable from the back of a room.

Artefact type	Accepted formats	Rules
Code	Git repository link plus a zipped snapshot	A README that explains how to run the work, a requirements or dependency file and no secrets committed.
Data outputs	.csv, .xlsx, .parquet or a database dump	Include a data dictionary for every output table.
Dashboards	.pbix, .twbx or a published link plus screenshots	Provide screenshots in a PDF in case the assessor cannot open the native file.
Recordings	A link to an unlisted video, maximum five minutes	Camera optional. Audio must be clear. Share settings must allow the Programme Office to view.

5.4 Deadlines and extensions

All deliverables are due by **Friday at 17:00 UK time** unless your track brief states an earlier interim deadline. Extensions are granted only for genuine reasons and must be requested before the deadline. Work submitted late without an agreed extension is marked as normal, but the overall band is capped at Meets expectations. Work submitted more than forty eight hours late is not assessed.

6. How You Will Be Assessed

6.1 The competency framework

Every track is assessed against the same five competencies so that employers can compare results across disciplines. Your track brief contains a rubric that interprets these competencies for your discipline and assigns weights to each one.

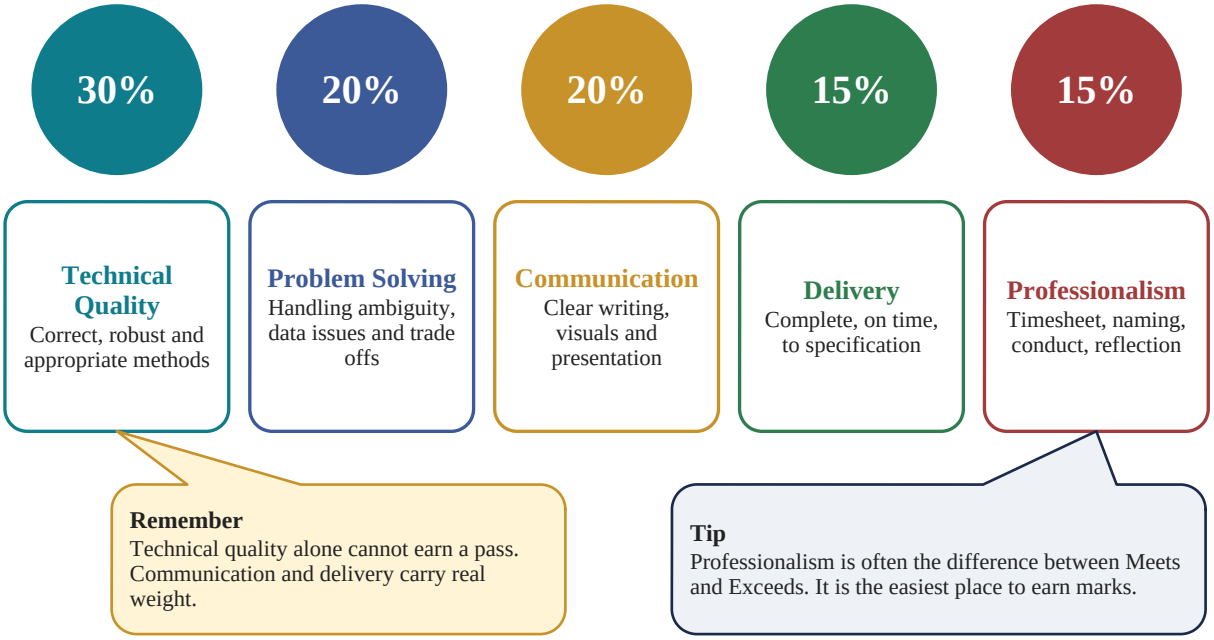


Figure 6.1. The five Vet2Employ competencies and their default weights

6.2 Assessment bands

Band	Description	Outcome
Exceeds expectations	Work that would be accepted by a real client with minimal changes. Shows independent judgement, anticipates stakeholder needs and goes beyond the minimum in a way that adds value.	Work Readiness Statement with commendation
Meets expectations	Work that satisfies every acceptance criterion with minor issues. Shows sound technique and clear communication.	Work Readiness Statement
Approaching expectations	Most deliverables are present and mostly correct, but one or more acceptance criteria are not met or the quality is inconsistent.	Detailed feedback and an invitation to resubmit within two weeks
Not yet meeting	Significant deliverables are missing, incorrect or not the participant's own work.	Detailed feedback and an invitation to join a later cohort

6.3 Feedback

You will receive written feedback on every deliverable within ten working days of the end of the week. Feedback follows a simple structure: what worked well, what would make it better and one priority action for your development. You may request a fifteen minute feedback call with your supervisor. Many participants tell us that this call is the most valuable part of the programme.

6.4 Track overview

The table below summarises the four tracks. Your track was confirmed in your joining email.

Track	Client (fictitious)	Core problem	Planned effort
1. Data Analyst	Kestrel Lane Home and Garden Ltd	Falling margin and fragmented reporting across 38 stores and an online shop	27.0 hours
2. Software Developer and Engineer	Northwold Couriers Ltd	No reliable parcel tracking; contact centre overwhelmed by status enquiries	26.0 hours
3. Cybersecurity	Ashbourne Valley Credit Union	Business email compromise and attempted payment fraud; weak security posture	27.0 hours
4. Data Engineering	Voltaway Charging Networks Ltd	Untrustworthy charging, revenue and reliability data spread across disconnected sources	26.0 hours

VET2EMPLOY | TRACK 1 OF 4

Data Analyst

1

Turn two years of fragmented retail data into one trusted view of performance, and tell the board where the margin went.

CLIENT ENGAGEMENT (FICTITIOUS)

Kestrel Lane Home and Garden Ltd

27	12	8	5
planned hours (minimum 20)	sequential activities	required deliverables	working days Monday to Friday

Note from your supervisor
The data has been made messy on purpose. Your quality log is assessed as carefully as your dashboard.

Your mission this week

Profile and clean seven messy retail datasets, build a single set of agreed KPIs, analyse sales, margin, promotions and loyalty behaviour, and deliver an interactive dashboard, an insight report and a ten minute executive presentation that the Kestrel Lane board can act on before its January planning round.

7. Track 1: Data Analyst

7.1 Welcome note from your Track Supervisor

Welcome to the Data Analyst track. This week you are joining a small Phexara analytics squad that has been engaged by Kestrel Lane Home and Garden, a family founded retailer with a loyal customer base and a growing problem. Sales are holding up on the surface, but profit is quietly slipping away and nobody in the business can agree on why. Every regional manager has their own spreadsheet, every spreadsheet tells a different story and the board has run out of patience.

Your job is to become the person in the room who can say, with evidence, what is actually happening. That means doing the unglamorous work first. You will profile the data, find the problems, fix what can be fixed, document what cannot and agree definitions before you calculate anything. Only then will you build the analysis and the dashboard. Analysts who skip straight to charts usually end up presenting confident answers to the wrong question, and you will not be one of them.

I will be looking for three things above everything else: that your numbers reconcile, that your assumptions are written down and that your recommendations are specific enough for a busy director to act on. If you get those three right, the rest will follow. Read the whole brief, plan your week on Monday and talk to me early if something does not add up. Good luck.

Your Track Supervisor, Data Analyst Track
Phexara Analytics Practice

7.2 The client: Kestrel Lane Home and Garden Ltd

Fact	Detail
Business	Independent home, garden and DIY retailer founded in 1994 in Warrington, Cheshire
Footprint	38 stores across the North West of England and the Midlands, plus an online store with click and collect
Size	Approximately 2,100 colleagues; annual turnover of about £186 million
Product range	Roughly 14,000 active product lines across 9 departments, from garden furniture and plants to paint, tools and seasonal goods
Loyalty scheme	Kestrel Rewards, launched in 2019, with about 410,000 registered members
Your sponsor (persona)	Priya Raman, Head of Commercial Insight. Practical, data literate and under pressure from the Finance Director to produce a single version of the truth
Other stakeholders	Daniel Hughes, Finance Director; Megan Lowe, Head of Marketing; Tom Brierley, Regional Operations Manager for the Midlands

7.3 The problem you are solving

Background

For most of its history Kestrel Lane grew steadily by opening new stores and building a reputation for helpful service. Over the last two years the business has faced rising costs, aggressive competition from national chains and online marketplaces, and a shift in customer behaviour towards smaller, more frequent baskets. In response, the Marketing team increased the number and depth of promotions, and the Commercial team widened the range of lower priced own label products.

Management accounts for the twelve months to September 2026 show that total revenue fell by only 1.9 percent compared with the previous year, which the board initially regarded as a good result in a difficult market. However, gross margin fell from 41.2 percent to 37.9 percent over the same period, which represents a loss of several million pounds of gross profit. Like for like sales, which exclude stores opened or closed during the period, fell by 4.8 percent. Footfall counters suggest that the number of store visits has been broadly stable, which implies that customers are still coming in but spending less, buying cheaper items or buying mainly on promotion.

What is going wrong

Priya Raman has summarised the situation in a briefing note that she shared with Phexara. Her main concerns are listed below.

- **There is no single version of the truth.** Each of the four regions produces its own weekly spreadsheet from exports of the point of sale system. Store codes, product categories and week definitions differ between regions, so totals never agree with the Finance figures.
- **Nobody knows whether promotions are working.** Marketing believes that promotions drive footfall and basket size. Finance believes that they mainly give away margin on items that customers would have bought anyway. Neither side has evidence.
- **The loyalty programme is under scrutiny.** Kestrel Rewards costs about £1.4 million a year to run. The number of members who shopped in the last ninety days appears to be falling, but the definition of an active member has never been agreed.
- **Returns are rising.** Store managers report more returns, particularly of garden furniture and power tools, but returns are recorded inconsistently and are sometimes netted off against sales.
- **Online and store data do not connect.** Online orders are held in a separate system, and click and collect orders may be counted twice: once online and again when collected in store.

Why it matters now

The board meets in January 2027 to approve the trading plan for the next financial year. The Finance Director has said that he will not approve another year of deep promotions without

evidence of their return, and the Head of Marketing has said that cutting promotions without evidence would be reckless. Priya needs an independent, evidence based view of performance that both sides will trust, and she needs it in a form that can be maintained after Phexara leaves.

The current data landscape

The diagram below shows how data currently flows from the operational systems to the people who use it. The annotations mark the weaknesses that you are expected to find and address during the week.

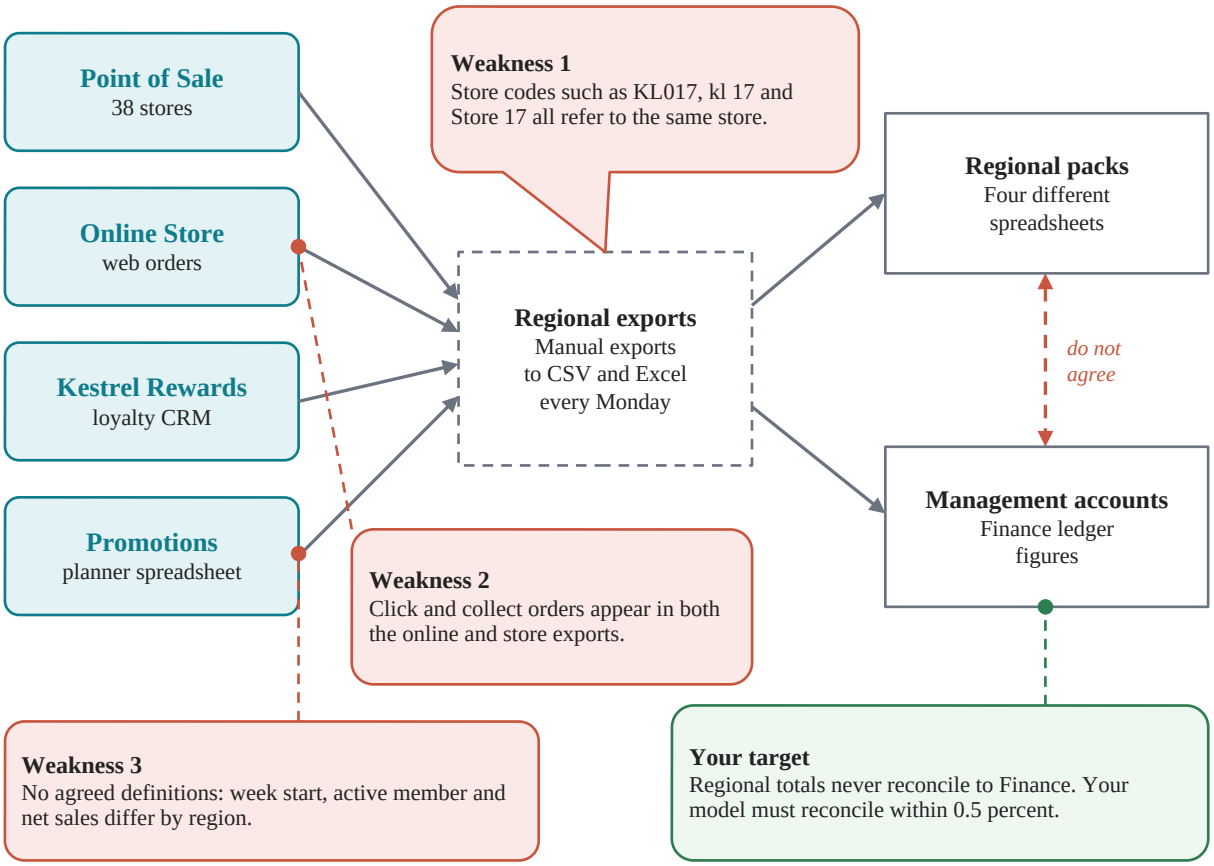


Figure 7.1. Current state data flow at Kestrel Lane, with known weaknesses annotated

What success looks like

By Friday Priya expects to have the following in her hands:

1. A cleaned, documented and reconciled analytical dataset that covers twenty four months of trading.
2. A short list of agreed KPI definitions that Finance and Marketing can both sign up to.
3. Clear answers, backed by analysis, to four business questions: Where has margin been lost? Do promotions generate incremental profit? Which customers are we losing? What is driving returns?
4. An interactive dashboard that regional managers can use every week without rebuilding spreadsheets.

5. Three to five specific, costed or sized recommendations that the board can debate in January.

Scope and constraints

In scope	Out of scope
Twenty four months of transactions from October 2024 to September 2026	Forecasting or demand planning models
Store, online, loyalty, promotions, products and returns data	Supplier performance and stock availability
Descriptive and diagnostic analysis, customer segmentation	Changes to the point of sale or online systems
A dashboard prototype and handover documentation	Staffing, payroll and HR data

- Use Excel, SQL and one business intelligence tool (Power BI or Tableau). Python or R may be used for cleaning and analysis but are not required.
- The dashboard must be usable by managers who are not analysts, on a standard laptop screen.
- Customer level data must be pseudonymised in any output. Do not display names or email addresses.
- All monetary values are in pounds sterling and include VAT unless stated otherwise in the data dictionary. You must decide and state whether you report gross or net of VAT.

7.4 Your working materials

The following files are provided in the materials folder for your track. Read the data dictionary before opening any file. Several issues have been planted in the data, and you should expect to find more than the dictionary admits.

File	Grain and size	Key fields	Known or suspected issues
kl_transactions.csv	One row per item line; about 1.2 million rows	txn_id, txn_datetime, store_code, product_id, qty, unit_price, discount_value, promo_code, member_id, channel	Duplicated rows, returns recorded as negative quantities, mixed date formats
kl_online_orders.csv	One row per order line; about 160,000 rows	order_id, order_date, fulfilment_type, collect_store, product_id, qty, price	Click and collect lines also appear in transactions
kl_products.xlsx	One row per product; about 14,600 rows	product_id, description, department, category, own_label_flag, unit_cost, launch_date	Missing unit costs, inconsistent category names
kl_stores.csv	One row per store; 41 rows	store_code, store_name, region, opening_date, closing_date, sales_area_sqft	Three closed stores, inconsistent code formats
kl_members.csv	One row per member; about 410,000 rows	member_id, join_date, home_postcode_area, age_band, marketing_consent	Test accounts, missing postcode areas

File	Grain and size	Key fields	Known or suspected issues
kl_promotions.xlsx	One row per promotion; about 520 rows	promo_code, mechanic, start_date, end_date, discount_depth, departments	Overlapping promotions, codes missing from transactions
kl_returns.csv	One row per return line; about 48,000 rows	return_id, original_txn_id, return_date, reason_code, qty, refund_value	Reason codes changed in April 2025
kl_data_dictionary.pdf	Reference document	Field descriptions and business rules	Out of date in places; treat with caution

Supervisor hint

Footfall data has not been provided. If you want to make a claim about conversion or footfall, you must request the data through the client question window and justify why you need it. Not every request will be granted.

7.5 Your week at a glance

No.	Activity	When	Effort
1	Induction, brief review and environment set up	Mon morning	1.5 hours
2	Stakeholder questions and analysis plan	Mon afternoon	2.0 hours
3	Data profiling and quality audit	Mon and Tue	3.0 hours
4	Data cleaning and transformation	Tue	3.0 hours
5	Data model and KPI definitions	Tue and Wed	2.0 hours
6	Sales and margin performance analysis	Wed	3.0 hours
7	Promotions effectiveness analysis	Wed and Thu	2.5 hours
8	Customer and loyalty segmentation	Thu	2.5 hours
9	Interactive dashboard build	Thu	3.0 hours
10	Insight report and recommendations	Fri morning	2.5 hours
11	Executive presentation and recording	Fri afternoon	1.0 hours
12	Reflection, timesheet and handover	Fri afternoon	1.0 hours
	Total planned effort	Monday to Friday	27.0 hours

7.6 Sequential activities

Complete the activities in the order given. Each one depends on the outputs of the previous ones. Where an activity spans two days, you may split it, but do not start an activity before the one it depends on is complete.

Activity 1: Induction, brief review and environment set up

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Monday morning	**1.5 hours**	**None**	**Ready environment**

Objective

Understand the engagement, confirm that you can access every material and tool, and set up a working folder that will keep your week organised.

What you must do, in order

1. Attend the induction call at 09:30 and note any clarifications given by your supervisor.
2. Read this entire track brief, including the deliverables and rubric, before opening any data.
3. Create your submission folder using the structure in Section 5.2 and copy the materials into a read only location inside 02_Working. Never edit the original files.
4. Install or confirm access to Excel, a SQL environment (for example SQLite, PostgreSQL or SQL Server Express) and either Power BI Desktop or Tableau Public.
5. Open each file once to confirm that it loads. Record row counts and file sizes in a simple inventory table.
6. Start your timesheet and your reflection log using the templates in the appendices.

You are done when

- Your folder structure matches Section 5.2.
- Every material file opens and its row count is recorded.
- Your timesheet has its first entry.

Activity 2: Stakeholder questions and analysis plan

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Monday afternoon	**2.0 hours**	**Activity 1**	**D1 Analysis Plan**

Objective

Translate the client's concerns into precise analytical questions, identify what you need to know before you start, and produce a plan that your supervisor can approve.

What you must do, in order

1. List each stakeholder named in Section 7.2 and write down what they want from this work and what they fear it might show.
2. Break each of the four business questions into two to four measurable sub questions. For example, Where has margin been lost? becomes: by department, by channel, by promotion status and by own label versus branded.
3. For every sub question, name the data fields you will use, the metric you will calculate and the chart or table you expect to produce.
4. Write between five and eight clarification questions for the client persona. Prioritise the questions that would change your approach if answered differently. Submit them before the question window closes on Tuesday at 12:00.

5. Draft a one page analysis plan (D1) that includes the questions, the approach, the assumptions you are making until the client replies and a day by day schedule.
6. Send D1 to your supervisor for approval by the end of Monday.

You are done when

- D1 is no longer than two pages and fits the template headings: Questions, Data, Method, Assumptions, Schedule, Risks.
- At least five clarification questions have been submitted.
- Your supervisor has acknowledged the plan.

> **What a good question looks like**
>
> Weak: Is the data correct? Strong: Should click and collect sales be attributed to the collecting store or to the online channel when reporting regional performance?

Activity 3: Data profiling and quality audit

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Monday and Tuesday	**3.0 hours**	**Activity 2**	**D2 Data Quality Log**

Objective

Measure the quality of every dataset systematically before changing anything, so that every later decision is grounded in evidence.

What you must do, in order

1. For each file, record the number of rows and columns, the data type of each field, the percentage of nulls, the number of distinct values and the minimum and maximum of numeric and date fields.
2. Check the uniqueness of every intended key (txn_id with product line, order_id, product_id, store_code, member_id, promo_code). Record any duplicates.
3. Check referential integrity: do all product_id values in transactions exist in the product table? Do all store_code values match the store table after standardisation? Do promo_code values match the promotions table?
4. Look for business rule violations: prices of zero or above a sensible ceiling, discounts larger than the price, transactions outside trading hours, sales at closed stores and member join dates after their first purchase.
5. Identify test or internal accounts in the member table, for example email domains or names that indicate staff testing. Use the pattern rather than personal data in your notes.
6. Record each issue in the Data Quality Log with an ID, dataset, field, description, number of affected rows, percentage of total, severity (High, Medium, Low) and a proposed treatment.
7. Summarise the five most material issues in a short paragraph at the top of the log.

You are done when

- Every dataset has a profile summary.
- At least twelve distinct issues are logged with counts and proposed treatments.
- Each High severity issue explains its potential effect on the business questions.

Activity 4: Data cleaning and transformation

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Tuesday	**3.0 hours**	**Activity 3**	**Clean tables and change log**

Objective

Apply the treatments from your quality log in a repeatable way, so that the same steps could be rerun next month on fresh data.

What you must do, in order

1. Choose your tool for cleaning (SQL scripts, Power Query or Python) and keep every step in code or in recorded query steps. Manual edits to cells are not permitted.
2. Standardise store codes to a single format and map historical variants using a lookup table.
3. Convert all dates to a single date type. Where a date is ambiguous, apply the rule agreed with the client or document your assumption.
4. Remove exact duplicate rows and record the number removed.
5. Separate returns from sales. Create a clear flag rather than allowing negative quantities to net off silently.
6. Identify click and collect orders that appear in both the online and store data, and decide on a single attribution rule. Record the rule in your decision log.
7. Impute or exclude missing unit costs using a documented method, such as the category average cost ratio, and flag every imputed value.
8. Reconcile your cleaned monthly net sales to the Finance totals supplied in the data dictionary. Investigate and explain any gap above 0.5 percent.
9. Update the Data Quality Log with the treatment applied and the rows affected for each issue.

You are done when

- Cleaning is fully reproducible from the raw files.
- Monthly totals reconcile to Finance within 0.5 percent, or the gap is explained.
- A change log lists every transformation in order.

Activity 5: Data model and KPI definitions

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Tuesday and Wednesday	**2.0 hours**	**Activity 4**	**D3 KPI Dictionary**

Objective

Design a simple, robust analytical model and agree precise definitions for the measures that everyone will use.

What you must do, in order

1. Design a star schema with a central sales fact table and dimensions for date, store, product, customer, promotion and channel. Draw it and annotate the grain of the fact table.
2. Build a date dimension that follows the Kestrel Lane trading week, which runs Sunday to Saturday, and includes financial year and period.
3. Write formal definitions for at least twelve KPIs, including gross sales, net sales, gross margin value, gross margin percentage, like for like sales, average transaction value, units per transaction, promotional participation rate, return rate, active member rate, member share of sales and ninety day retention.
4. For each KPI record: name, business definition, formula, inclusions, exclusions, owner, refresh frequency and an example calculation.
5. Load the model into your BI tool and confirm that relationships, filters and totals behave as expected.

You are done when

- D3 contains at least twelve KPIs in the required format.
- The model diagram shows grain, keys and relationships.
- A test calculation for each KPI matches a manual check.

Activity 6: Sales and margin performance analysis

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Wednesday	**3.0 hours**	**Activity 5**	**Analysis workbook section A**

Objective

Find where gross margin has been lost and quantify each driver, so that the board knows which levers matter most.

What you must do, in order

1. Produce a monthly trend of net sales and gross margin percentage for both years, split by channel.
2. Build a margin bridge that explains the change in gross profit between the two years using volume, price, mix and promotion effects. State your method clearly.
3. Rank departments and categories by their contribution to the margin decline.

4. Compare own label with branded products, and new lines launched in the last year with established lines.
5. Calculate like for like sales by store and region, excluding stores that opened or closed in the period, and identify the five strongest and five weakest stores.
6. Write three to five headline findings, each supported by a figure and a chart.

You are done when

- The margin bridge reconciles to the total change in gross profit.
- Findings are quantified in pounds and percentage points, not just described.
- Every chart has a title that states the finding, not just the subject.

Activity 7: Promotions effectiveness analysis

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Wednesday and Thursday	**2.5 hours**	**Activity 6**	**Analysis workbook section B**

Objective

Estimate whether promotions generate incremental profit or simply give margin away.

What you must do, in order

1. Classify each promotion by mechanic (percentage off, multibuy, bundle, loyalty only) and by discount depth band.
2. For a sample of at least twenty promotions, compare sales of promoted products during the promotion with a baseline from the four comparable weeks before it.
3. Estimate uplift in units and in gross profit, and look for evidence of pull forward (lower sales after the promotion) and cannibalisation of similar products.
4. Identify the promotions that destroyed the most profit and those that generated the most incremental profit. Describe what the best performers have in common.
5. State the limitations of your method honestly. A baseline comparison is not a controlled experiment.

You are done when

- At least twenty promotions are evaluated using a consistent method.
- Results distinguish between sales uplift and profit uplift.
- Limitations are stated clearly and fairly.

Activity 8: Customer and loyalty segmentation

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Thursday	**2.5 hours**	**Activity 7**	**Analysis workbook section C**

Objective

Understand which members are drifting away and which are most valuable, using a transparent segmentation that marketing can use.

What you must do, in order

1. Exclude test accounts and agree a definition of an active member with reference to your KPI dictionary.
2. Calculate recency, frequency and monetary value for each member over the last twelve months and assign scores from one to five for each.
3. Group members into between five and seven named segments, such as Champions, Loyal Regulars, Promotion Seekers, At Risk and Lapsed. Document the rules for each.
4. Profile each segment by size, share of sales, margin, promotion dependency, preferred departments and region.
5. Calculate ninety day retention by joining cohort and identify when members are most likely to lapse.
6. Estimate the value at risk if the At Risk segment lapses fully.

You are done when

- Segment rules are documented and reproducible.
- No personal data appears in outputs.
- Value at risk is quantified with stated assumptions.

Target dashboard layout

Before you build the dashboard in Activity 9, study the wireframe below. It shows the layout that Priya Raman has asked for. You may improve on it, but you must justify any change in your decision log.

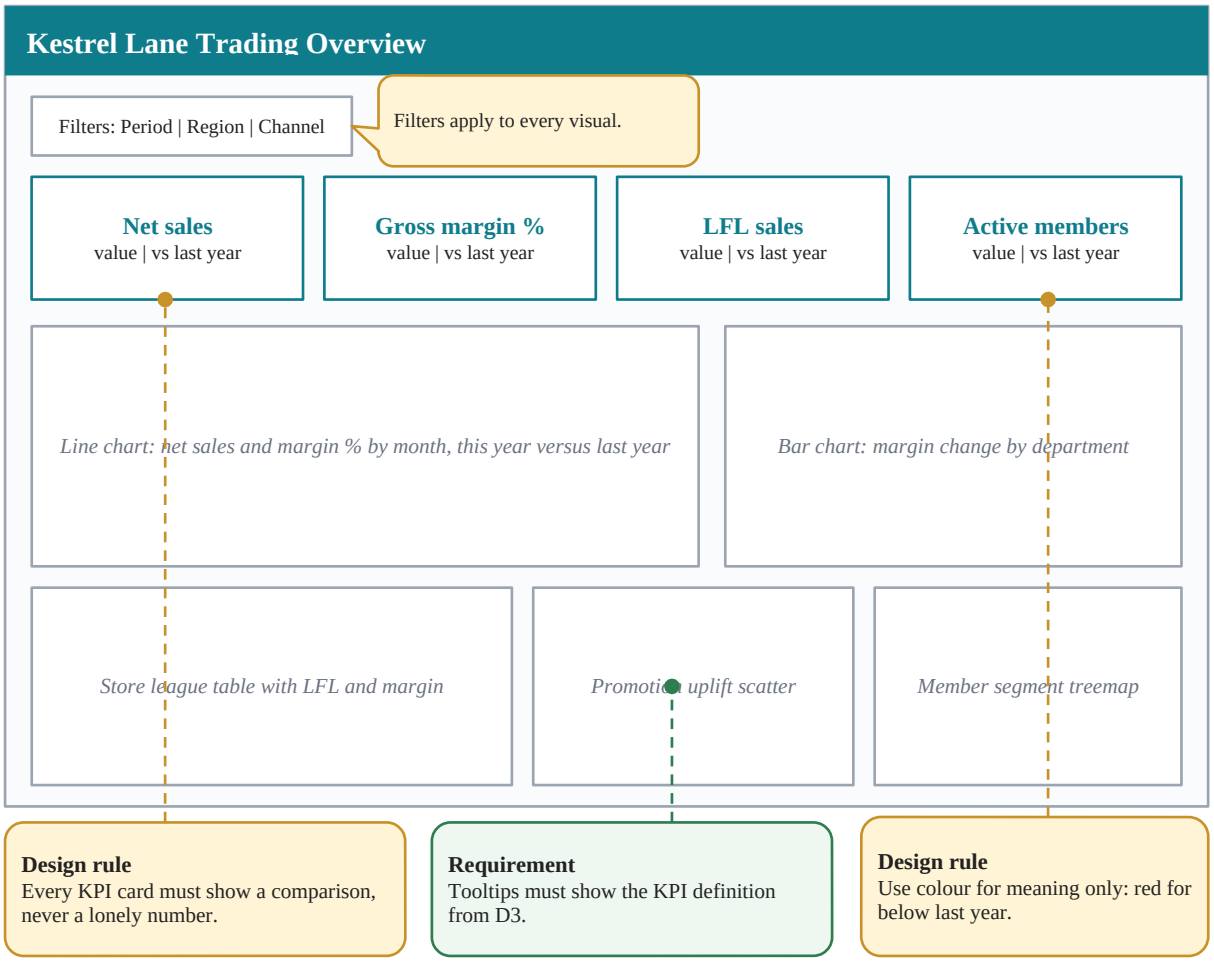


Figure 7.2. Dashboard wireframe requested by the client, annotated with design requirements

Activity 9: Interactive dashboard build

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Thursday	**3.0 hours**	**Activities 5 to 8**	**D4 Dashboard**

Objective

Build a dashboard that lets regional managers answer their own weekly questions without rebuilding spreadsheets.

What you must do, in order

1. Sketch your layout on paper first, starting from the wireframe in Figure 7.2.
2. Build the overview page with KPI cards, trends and comparisons to last year.
3. Add a second page for promotions and a third page for customers and segments.
4. Add filters for period, region, store, channel and department, and make sure that they interact correctly.
5. Apply accessible design: sufficient contrast, readable fonts, no reliance on colour alone and clear titles.
6. Test every number on the dashboard against your workbook and record the test in a short checklist.

7. Ask your peer reviewer to complete three tasks on the dashboard without help and note where they struggled. Fix the biggest problem.

You are done when

- The dashboard has at least three pages and the filters work on every visual.
- Every KPI matches the workbook and the KPI dictionary.
- A peer usability test has been completed and one improvement made.

Activity 10: Insight report and recommendations

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Friday morning	**2.5 hours**	**Activity 9**	**D5 Insight Report**

Objective

Write a concise report that turns your analysis into decisions for the board.

What you must do, in order

1. Start with a one page executive summary that answers the four business questions directly.
2. Follow with sections on margin, promotions, customers and returns, each with no more than three charts.
3. Make between three and five recommendations. For each one, state the action, the owner, the expected impact in pounds or percentage points, the evidence and the risk.
4. Include a section on data quality that explains the most important issues, what you did about them and what the business should fix at source.
5. Add appendices for methods, KPI definitions and assumptions.
6. Proofread the report and check every figure against the workbook.

You are done when

- The report is no longer than twelve pages excluding appendices.
- Every recommendation is specific, owned and sized.
- Every figure in the report can be traced to the workbook.

Activity 11: Executive presentation and recording

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Friday afternoon	**1.0 hours**	**Activity 10**	**D6 Presentation**

Objective

Present your findings to the board in ten minutes, as you would in a real client meeting.

What you must do, in order

1. Build no more than ten slides. Use one message per slide and an action title for each.
2. Structure the story as situation, complication, findings and recommendations.
3. Rehearse twice with a timer.
4. Record a five minute version and upload it as an unlisted link.

5. Prepare answers to the three hardest questions you expect from the Finance Director and the Head of Marketing.

You are done when

- The deck has ten slides or fewer with action titles.
- The recording is five minutes or less and clearly audible.
- Three anticipated questions and answers are written in the speaker notes.

Activity 12: Reflection, timesheet and handover

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Friday afternoon	**1.0 hours**	**Activity 11**	**D7 Handover Note and D8 Reflection**

Objective

Make your work maintainable after you leave and reflect honestly on what you learned.

What you must do, in order

1. Write a handover note that explains how to refresh the data, rerun the cleaning steps and update the dashboard each month.
2. List known limitations and recommended next steps for the analytics team.
3. Complete your timesheet and check that it totals at least twenty hours.
4. Complete your reflection log using the template in Appendix C, including your AI use declaration.
5. Check every file against the naming convention and upload to your submission folder by 17:00.

You are done when

- Someone else could refresh the dashboard using only your handover note.
- Timesheet and reflection log are complete.
- All deliverables are uploaded and correctly named.

7.7 Required deliverables

Ref	Deliverable	Format and file name	Due	Acceptance criteria
D1	Analysis Plan	PDF, V2E_DA_[Surname]_D1_AnalysisPlan_v1.pdf	Mon 17:00	Questions, data, method, assumptions, schedule and risks; approved by supervisor
D2	Data Quality Log	XLSX, V2E_DA_[Surname]_D2_DataQualityLog_v1.xlsx	Tue 17:00	At least twelve issues with counts, severity, treatment and impact
D3	KPI Dictionary and Data Model	XLSX and PNG diagram	Wed 12:00	At least twelve KPIs with full definitions; annotated star schema
D4	Interactive Dashboard	PBIX or TWBX plus screenshot PDF	Fri 17:00	Three pages, working filters, reconciled numbers, accessible design

Ref	Deliverable	Format and file name	Due	Acceptance criteria
D5	Insight Report	PDF and DOCX	Fri 17:00	Executive summary, four answered questions, three to five sized recommendations
D6	Executive Presentation	PDF, PPTX and video link	Fri 17:00	Ten slides or fewer, action titles, five minute recording
D7	Handover Note and Working Files	PDF plus scripts, queries and workbook	Fri 17:00	Reproducible cleaning steps and refresh instructions
D8	Timesheet and Reflection Log	XLSX and PDF	Fri 17:00	At least twenty hours logged; AI use declared; decision log complete

Where each deliverable goes

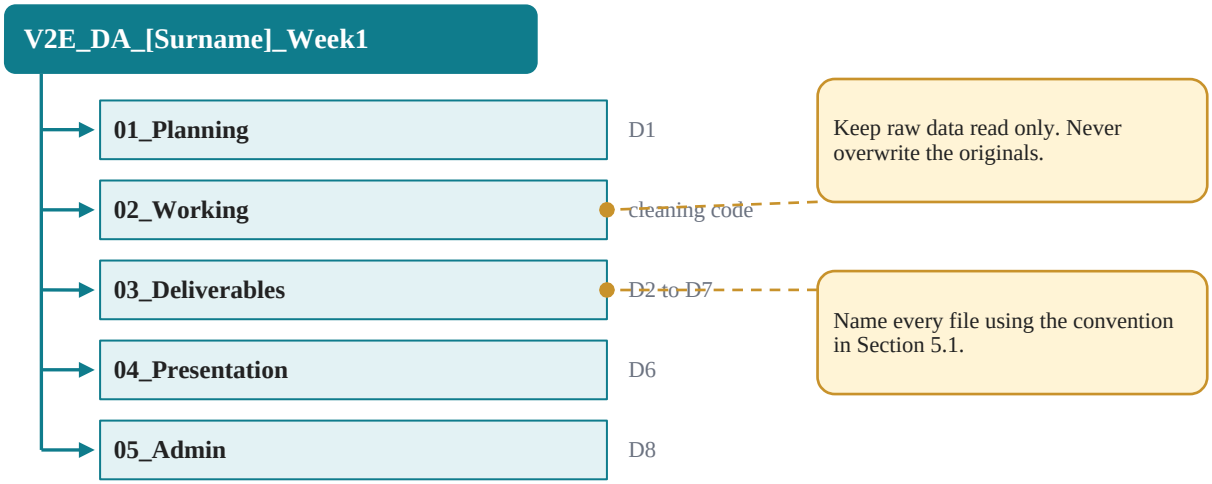


Figure 7.3. Data Analyst submission folder with deliverable placement

7.8 Assessment rubric

Criterion	Weight	Exceeds expectations	Meets expectations	Not yet meeting
Technical quality	30%	Reproducible cleaning, reconciled model, sound and clearly explained methods for margin bridge, uplift and segmentation	Correct calculations, mostly reproducible, methods explained	Calculations do not reconcile or cannot be reproduced
Problem solving	20%	Finds planted and unplanted issues, makes justified decisions and quantifies their effect	Finds most planted issues and documents treatments	Issues ignored or treated without explanation
Communication	20%	Report and slides tell a compelling, evidence based story with action titles and sized recommendations	Clear report and slides with relevant charts	Charts without insight; recommendations vague
Delivery	15%	All deliverables complete, on time and polished	All deliverables complete and on time	Missing or late deliverables
Professionalism	15%	Exemplary naming, timesheet, decision log and peer feedback	Conventions followed; timesheet and reflection complete	Conventions ignored; timesheet missing

7.9 Stretch goals

If you complete every activity early, choose one of the stretch goals below and record the time spent in your timesheet.

- Build a simple returns analysis that links return reasons to products, stores and promotions, and estimate the margin cost of returns by department.
- Use a basket analysis (association rules) to identify product pairs that are frequently bought together, and suggest bundle promotions that protect margin.
- Create a simple statistical test or confidence interval for your promotion uplift estimates.
- Write a short data governance proposal for Kestrel Lane covering data ownership, a master store list and a KPI change process.

VET2EMPLOY | TRACK 2 OF 4

Software Developer and Engineer

2

Design, build, test and document a parcel booking and tracking service that stops customers having to phone to ask where their parcel is.

CLIENT ENGAGEMENT (FICTITIOUS)

Northwold Couriers Ltd

26	12	8	5
planned hours (minimum 20)	sequential activities	required deliverables	working days Monday to Friday

Note from your supervisor
Working, tested and documented beats clever and fragile. Commit little and often.

Your mission this week

Deliver a working minimum viable product called Northwold Track: a tested web API with a data model, a validated parcel status lifecycle, a public tracking page, protected staff and driver endpoints, automated tests in a continuous integration pipeline and the documentation another engineer would need to pick it up on Monday.

8. Track 2: Software Developer and Engineer

8.1 Welcome note from your Track Supervisor

Welcome to the Software Developer and Engineer track. This week you are a member of a small Phexara product engineering team that has been asked to help Northwold Couriers, a regional parcel carrier whose customers are frustrated and whose contact centre is drowning in phone calls. The client does not need a perfect system by Friday. It needs a credible, working first version that proves the idea, that its own developers can extend and that does not fall over the first time a driver scans a parcel twice.

You will be working the way professional teams work. You will turn a vague request into user stories with acceptance criteria, design before you code, use version control properly, write tests that actually catch defects, review your own work for security problems and write documentation for the next person. I care much more about how you work than about which language or framework you choose. A small, well structured, well tested service will score far higher than a large, untested one with every feature half finished.

Commit your work little and often with meaningful messages, keep your main branch working at all times and tell me early if you think the scope is too large for the week. Negotiating scope is part of the job. I look forward to seeing your first pull request.

Your Track Supervisor, Software Engineering Track
Phexara Product Engineering

8.2 The client: Northwold Couriers Ltd

Fact	Detail
Business	Regional parcel and pallet courier founded in 2008, headquartered in Wakefield, West Yorkshire
Operations	Two sorting depots (Wakefield and Gateshead), about 140 vans and 190 drivers covering Yorkshire, the North East and Cumbria
Volume	About 9,000 parcels a day, rising to about 15,000 a day between November and January
Customers	About 600 business senders, mainly small online retailers, and the recipients of their parcels
Your sponsor (persona)	Gareth Whitfield, Operations and Technology Director. Ex driver, pragmatic, sceptical of big IT projects after a failed system replacement in 2022
Other stakeholders	Leanne Ashworth, Customer Service Manager; Sam Iqbal, the company's only developer; Fiona Clarke, Depot Manager at Gateshead

8.3 The problem you are solving

Background

Northwold grew quickly during the expansion of online shopping by offering reliable next day delivery and friendly drivers at a lower price than national carriers. Its technology, however, has not kept pace. Parcels are booked by business customers through a web form that emails the depot. Depot staff re type each booking into a spreadsheet. Drivers carry handheld scanners that store scans locally and upload them in a batch file when the van returns to the depot at the end of the day. There is no system that tells anyone, in real time, where a parcel is.

What is going wrong

- **The contact centre is overwhelmed.** Leanne Ashworth's team of eleven receives about 1,800 calls a day, and she estimates that around 70 percent are customers asking where their parcel is. Average waiting time has reached fourteen minutes.
- **Business customers are leaving.** Two of the twenty largest senders moved to a competitor this year and both cited the lack of online tracking as the main reason.
- **Data is entered twice and errors creep in.** Re keying bookings causes wrong postcodes and missing phone numbers, which lead to failed deliveries that cost about £6 each to reattempt.
- **Status information is unreliable.** Because scans arrive in a nightly batch, the depot cannot see failed delivery attempts until the next morning, by which time the customer has already called.
- **The in house developer is stretched.** Sam Iqbal maintains the website, the scanner batch import and the payroll export. Anything built must be simple for him to understand, run and extend.

What the client has asked for

Gareth Whitfield wants a minimum viable product, called **Northwold Track**, that can be piloted at the Gateshead depot in the new year. In his own words from the briefing call: he wants senders to book parcels without anyone retyping them, drivers to update status from the road, and customers to type in a tracking number and see where their parcel is without picking up the phone. He has also made it clear that he does not want a system that nobody in the company can maintain.

Target architecture

The diagram below shows the target architecture that Phexara proposed and the client accepted in principle. You may refine it in Activity 3, but any significant change must be justified in an architecture decision record.

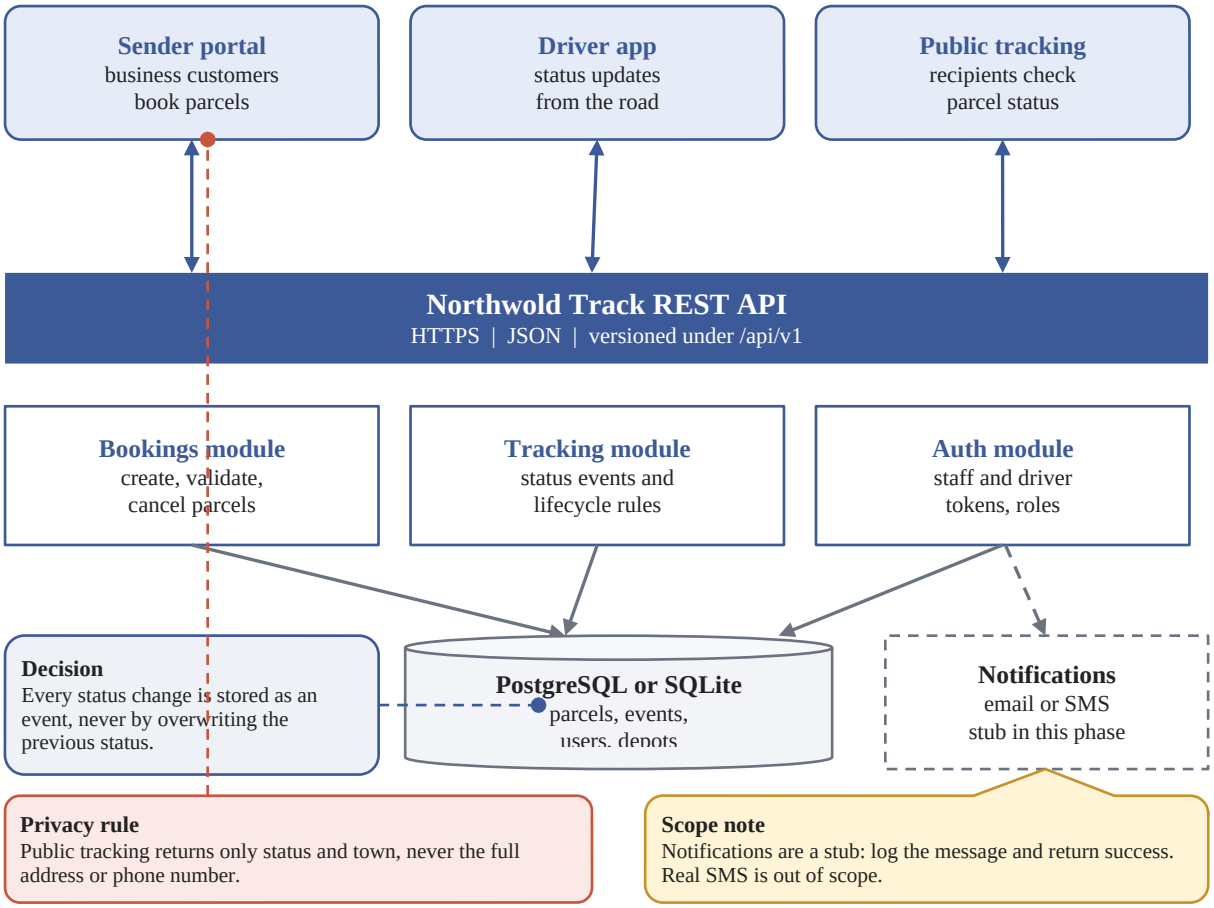


Figure 8.1. Northwold Track target architecture, annotated with design decisions

Parcel status lifecycle

A parcel moves through a defined sequence of statuses. The rules below come from the depot managers and are the most important business logic in the system. Your service must reject any transition that is not shown by an arrow.

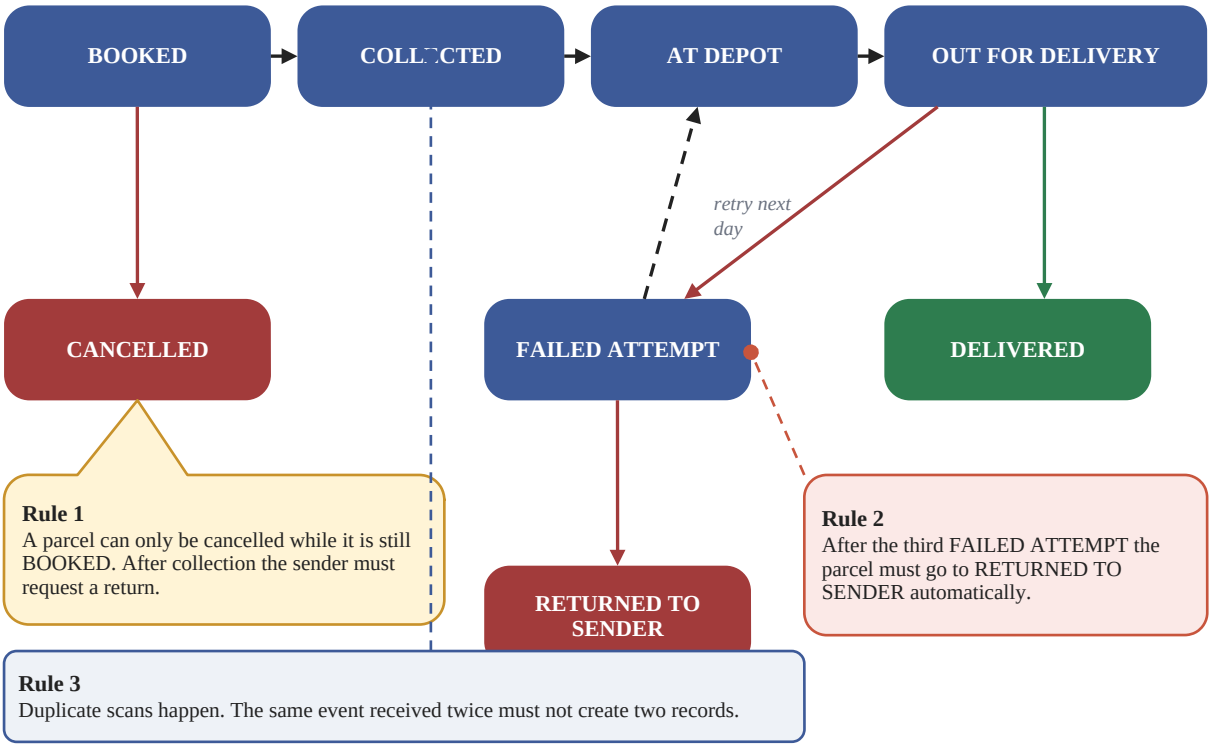


Figure 8.2. Permitted parcel status transitions with annotated business rules

Scope and constraints

Must have (in scope)	Out of scope this week
Create, read and cancel parcel bookings through the API	Payments, invoicing and pricing
Record status events and enforce the lifecycle rules	Route optimisation and driver scheduling
Public tracking by tracking number with privacy rules	A native mobile app (a simple web form or API calls are acceptable)
Role based access for staff and drivers	Integration with the real scanners
Automated tests and a continuous integration pipeline	Production hosting and real SMS or email

- Use one mainstream back end stack: Python with FastAPI or Django, JavaScript or TypeScript with Node.js and Express, Java with Spring Boot, or C# with ASP.NET Core. Justify your choice briefly.
- Use a relational database. SQLite is acceptable for the pilot provided that you explain how to move to PostgreSQL.
- The front end for public tracking can be simple server rendered HTML or a small single page application. It must be usable on a mobile phone.
- Use Git from the first hour, with a hosted repository (GitHub, GitLab or Bitbucket) and a continuous integration workflow that runs your tests on every push.
- No secrets may be committed to the repository. Use environment variables and provide an example configuration file.

8.4 Your working materials

Material	Description	How to use it
nw_requirements_notes.pdf	Notes from the briefing call with Gareth, Leanne and Fiona, including some contradictory statements	Extract requirements and flag contradictions as questions
nw_sample_bookings.csv	About 2,000 historic bookings with sender, recipient, postcode, weight and service level; includes invalid postcodes and missing fields	Seed data and test cases for validation rules
nw_scan_batch_sample.json	A day of driver scan events with duplicates, out of order timestamps and one scanner with a clock set one hour behind	Test data for event processing and idempotency
nw_depots.csv	Depot codes, names, addresses and the postcode areas each depot serves	Reference data for routing parcels to depots
nw_api_style_guide.pdf	Phexara house style for REST APIs: naming, error format, pagination and versioning	Follow it for all endpoints
nw_tracking_page_sketch.png	A rough sketch by Leanne of what customers should see	Starting point for the tracking page design

Supervisor hint

The requirements notes contain at least three statements that contradict each other or the lifecycle diagram. Finding them and resolving them through the client question window is part of the assessment. Do not silently pick one.

8.5 Your week at a glance

No.	Activity	When	Effort
1	Induction, brief review and tooling set up	Mon morning	1.0 hours
2	Requirements analysis and user stories	Mon	2.0 hours
3	System design: data model, API contract and decisions	Mon and Tue	3.0 hours
4	Repository, branching strategy and CI skeleton	Tue morning	1.5 hours
5	Data layer and parcel booking endpoints	Tue	3.0 hours
6	Status events and lifecycle rules	Wed	3.0 hours
7	Public tracking endpoint and tracking page	Wed and Thu	3.0 hours
8	Authentication and role based access	Thu	2.0 hours
9	Automated testing and coverage	Thu	3.0 hours
10	Security, logging and code quality review	Fri morning	1.5 hours
11	Documentation and API specification	Fri morning	1.5 hours
12	Release, demo recording and reflection	Fri afternoon	1.5 hours
	Total planned effort	Monday to Friday	26.0 hours

8.6 Sequential activities

Complete the activities in the order given. Although some development work can overlap in real projects, this week you must have your design approved (Activity 3) before writing feature code, and your continuous integration pipeline running (Activity 4) before you merge any feature.

Activity 1: Induction, brief review and tooling set up

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Monday morning	**1.0 hours**	**None**	**Working toolchain**

Objective

Understand the engagement and prove that your development environment works before you commit to a design.

What you must do, in order

1. Attend the induction call and record any clarifications.
2. Read this entire track brief, the requirements notes and the API style guide.
3. Install your chosen language runtime, a code editor, Git, a database client and an API client such as Postman, Insomnia or curl.
4. Create a hello world service in your chosen framework and run it locally to confirm that the toolchain works.
5. Create your submission folder and start your timesheet and reflection log.

You are done when

- A minimal service runs locally and responds to a request.
- Your timesheet and reflection log are started.

Activity 2: Requirements analysis and user stories

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Monday	**2.0 hours**	**Activity 1**	**D1 Requirements and Backlog**

Objective

Turn the client's informal requests into a prioritised, testable backlog.

What you must do, in order

1. Identify the user roles: business sender, recipient, driver, depot staff and administrator.
2. Write at least twelve user stories in the form As a role, I want a capability, so that a benefit. Cover booking, tracking, status updates, cancellation, access control and reporting.
3. Add acceptance criteria to every story in Given, When, Then form. For example: Given a parcel in status BOOKED, when the sender cancels it, then its status becomes CANCELLED and a cancellation event is recorded.

4. Prioritise the backlog using MoSCoW (Must, Should, Could, Won't) and estimate each story in hours or story points.
5. List the non functional requirements: response time, availability, privacy, security, accessibility and maintainability. Make each one measurable.
6. Record every contradiction you find in the requirements notes and submit clarification questions before the window closes.

You are done when

- At least twelve stories, each with acceptance criteria and a priority.
- At least six measurable non functional requirements.
- Contradictions are listed and questions submitted.

> **Measurable, not vague**
> Weak: The tracking page should be fast. Strong: The tracking endpoint returns a response within 300 milliseconds for 95 percent of requests with 10,000 parcels in the database.

Activity 3: System design: data model, API contract and decisions

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Monday and Tuesday	**3.0 hours**	**Activity 2**	**D2 Design Pack**

Objective

Design the service on paper so that the build is fast, consistent and reviewable.

What you must do, in order

1. Draw an entity relationship diagram for at least these entities: Sender, Parcel, StatusEvent, Depot and User. Show keys, relationships, data types and constraints.
2. Decide how you will generate tracking numbers. They must be unique, hard to guess and easy to read over the phone. Justify the format.
3. Specify every endpoint in a table: method, path, purpose, required role, request body, success response and error responses. Follow the house style guide, including the standard error format.
4. Write the lifecycle rules from Figure 8.2 as a transition table that your code will use directly.
5. Write at least three architecture decision records (ADRs), each with context, decision, alternatives considered and consequences. Suggested topics: choice of stack, choice of database, event sourcing of status history and approach to idempotency.
6. Review the design with your supervisor at the midweek review or earlier by message.

You are done when

- The design pack contains the ERD, endpoint table, transition table and three ADRs.
- Every user story marked Must is traceable to at least one endpoint.
- Your supervisor has seen the design before feature code is merged.

Activity 4: Repository, branching strategy and CI skeleton

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Tuesday morning	**1.5 hours**	**Activity 3**	**Repository with green pipeline**

Objective

Set up the engineering foundations so that every later change is safe, reviewed and tested.

What you must do, in order

1. Create a hosted repository named northwold_track with a README, a licence file and a .gitignore suitable for your stack.
2. Choose and document a branching strategy, such as trunk based development with short lived feature branches, and protect the main branch.
3. Add a formatter and a linter and configure them to run locally.
4. Create a continuous integration workflow (for example GitHub Actions) that installs dependencies, runs the linter and runs the test suite on every push and pull request.
5. Add a single placeholder test so that the pipeline runs green.
6. Add an example environment file and load configuration from environment variables.

You are done when

- The pipeline runs on every push and is green.
- No secrets are present anywhere in the repository history.
- The README explains how to run the service and the tests.

Activity 5: Data layer and parcel booking endpoints

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Tuesday	**3.0 hours**	**Activity 4**	**Booking feature merged**

Objective

Implement the core booking capability with robust validation.

What you must do, in order

1. Implement the database schema using migrations rather than manual SQL so that the schema can be recreated reliably.
2. Write a seed script that loads the depot reference data and a sample of valid bookings.
3. Implement endpoints to create a parcel, retrieve a parcel by tracking number for staff, list parcels with pagination and filtering by depot and status, and cancel a parcel.
4. Validate input: UK postcode format, required fields, weight limits by service level and phone number format. Return errors in the house style format with clear messages.
5. Assign the destination depot automatically from the recipient postcode area.
6. Record a BOOKED status event when a parcel is created.
7. Open a pull request, review your own diff carefully and merge only when the pipeline is green.

You are done when

- Valid bookings are stored and invalid bookings are rejected with helpful errors.
- Pagination works and is documented.
- The cancellation rule from Figure 8.2 is enforced.

Activity 6: Status events and lifecycle rules

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Wednesday	**3.0 hours**	**Activity 5**	**Tracking events feature merged**

Objective

Implement the business logic that matters most to the client, so that the status of every parcel is trustworthy.

What you must do, in order

1. Implement an endpoint for drivers and depot staff to post a status event with tracking number, new status, timestamp, location and an optional note.
2. Enforce the transition table. Reject invalid transitions with a clear error and do not change the parcel.
3. Make the endpoint idempotent. Use an event identifier supplied by the client device so that a repeated event is acknowledged but not stored twice.
4. Handle out of order events using the event timestamp, and decide how to treat events from the scanner whose clock is wrong. Record your decision in an ADR.
5. Implement Rule 2: the third failed attempt moves the parcel to RETURNED TO SENDER automatically.
6. Implement a batch import command that processes nw_scan_batch_sample.json through the same logic and reports how many events were accepted, rejected and ignored as duplicates.

You are done when

- Invalid transitions are rejected and the parcel is unchanged.
- Duplicate events do not create duplicate records.
- The batch import produces a clear summary for the sample file.

Activity 7: Public tracking endpoint and tracking page

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Wednesday and Thursday	**3.0 hours**	**Activity 6**	**Tracking page feature merged**

Objective

Give recipients a simple way to see where their parcel is, without exposing personal data.

What you must do, in order

1. Implement a public endpoint that returns, for a tracking number, the current status, an estimated delivery date and a timeline of events showing status, time and town only.

2. Apply the privacy rule: never return addresses, phone numbers or names through the public endpoint.
3. Protect the endpoint against enumeration: rate limit requests and return the same response for a tracking number that does not exist as for one that is malformed.
4. Build a tracking page with a single input, clear status display and a timeline. Make it responsive on a mobile phone and accessible to keyboard and screen reader users.
5. Write friendly, plain English status messages, such as Your parcel is out for delivery today.
6. Ask your peer reviewer to look up three parcels on a phone and record their feedback.

You are done when

- The page works on a narrow mobile screen.
- No personal data is visible in the public response.
- Rate limiting is demonstrated in a test or a recorded request sequence.

Activity 8: Authentication and role based access

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Thursday	**2.0 hours**	**Activity 7**	**Auth feature merged**

Objective

Make sure that only the right people can perform each action.

What you must do, in order

1. Implement user accounts for staff and drivers with securely hashed passwords using a modern algorithm such as bcrypt or Argon2.
2. Issue signed tokens (for example JSON Web Tokens) with a short expiry, and include the user role.
3. Apply role checks to every protected endpoint according to your endpoint table. Senders may only see their own parcels.
4. Return 401 for missing or invalid credentials and 403 for valid credentials without permission.
5. Write tests that prove a driver cannot cancel a booking and a sender cannot read another sender's parcels.

You are done when

- Every protected endpoint has an access test.
- Passwords are never stored or logged in plain text.

Activity 9: Automated testing and coverage

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Thursday	**3.0 hours**	**Activity 8**	**D4 Test Report**

Objective

Build confidence that the service works and will keep working as Sam extends it.

What you must do, in order

1. Write unit tests for validation, tracking number generation and every row of the transition table, including invalid transitions.
2. Write integration tests that call the API against a test database and cover the main user journeys end to end.
3. Write tests for idempotency, out of order events and the third failed attempt rule.
4. Measure test coverage and aim for at least 80 percent on the business logic modules. Explain any gaps.
5. Make sure that all tests run in the continuous integration pipeline.
6. Write a short test report: approach, what is covered, results, known defects and what you would test next.

You are done when

- The pipeline runs all tests and is green.
- Coverage on business logic is at least 80 percent or the shortfall is explained.
- The test report lists known defects honestly.

Activity 10: Security, logging and code quality review

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Friday morning	**1.5 hours**	**Activity 9**	**D5 Review Checklist**

Objective

Review your own service as a sceptical senior engineer would, and fix what matters most.

What you must do, in order

1. Review the service against the OWASP Top 10 for web applications and the OWASP API Security Top 10. Record each item as addressed, partly addressed or not applicable with a reason.
2. Check that inputs are validated, queries are parameterised and error messages do not leak internal details.
3. Add structured logging for requests and status changes, without logging personal data or tokens.
4. Run a dependency vulnerability scan using the tool for your ecosystem and record the results.
5. Fix the most important issues and record the rest as backlog items.

You are done when

- The checklist is complete with evidence.
- No high severity dependency vulnerability remains unexplained.

Activity 11: Documentation and API specification

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Friday morning	**1.5 hours**	**Activity 10**	**D3 API Specification and D6 Engineering Guide**

Objective

Make the service understandable and maintainable by someone who was not in the room.

What you must do, in order

1. Produce an OpenAPI specification for every endpoint, generated by your framework or written by hand, and check that it matches the implementation.
2. Update the README with an overview, an architecture diagram, set up steps, configuration, how to run tests and how to load seed data.
3. Write an engineering guide for Sam that explains the code structure, the lifecycle rules, how to add a new status safely and how to move from SQLite to PostgreSQL if you used SQLite.
4. List known limitations and a recommended roadmap for the next three months.

You are done when

- A new developer can run the service and tests from the README alone.
- The OpenAPI specification is complete and accurate.

Activity 12: Release, demo recording and reflection

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Friday afternoon	**1.5 hours**	**Activity 11**	**D7 Demo and D8 Reflection**

Objective

Ship a tagged release and demonstrate it as you would to the client.

What you must do, in order

1. Tag a release called v0.1.0 in your repository with release notes listing features, known issues and set up changes.
2. Record a demonstration of no more than five minutes: book a parcel, post status events, show an invalid transition being rejected, show the public tracking page and show the green pipeline.
3. Complete your timesheet, reflection log and AI use declaration.
4. Check names and upload every deliverable by 17:00.

You are done when

- A tagged release exists with release notes.
- The demonstration covers every Must have story.
- Timesheet and reflection log are complete.

8.7 Required deliverables

Ref	Deliverable	Format and file name	Due	Acceptance criteria
D1	Requirements and Backlog	PDF or XLSX, V2E_SE_[Surname]_D1_Backlog_v1	Mon 17:00	Twelve or more stories with acceptance criteria, priorities, estimates and NFRs
D2	Design Pack	PDF, V2E_SE_[Surname]_D2_DesignPack_v1.pdf	Tue 12:00	ERD, endpoint table, transition table and three ADRs
D3	Source Code and API Specification	Repository link, zipped snapshot and openapi file	Fri 17:00	Runs from the README, green pipeline, tagged v0.1.0, OpenAPI matches implementation
D4	Test Report	PDF, V2E_SE_[Surname]_D4_TestReport_v1.pdf	Fri 17:00	Approach, coverage, results and known defects
D5	Security and Quality Checklist	XLSX or PDF	Fri 17:00	OWASP review with evidence and dependency scan results
D6	Engineering Guide	PDF and Markdown in the repository	Fri 17:00	Structure, lifecycle rules, extension guide and roadmap
D7	Demonstration Recording	Video link, maximum five minutes	Fri 17:00	Covers every Must have story and shows the pipeline
D8	Timesheet and Reflection Log	XLSX and PDF	Fri 17:00	At least twenty hours logged; AI use declared; decision log complete

Where each deliverable goes

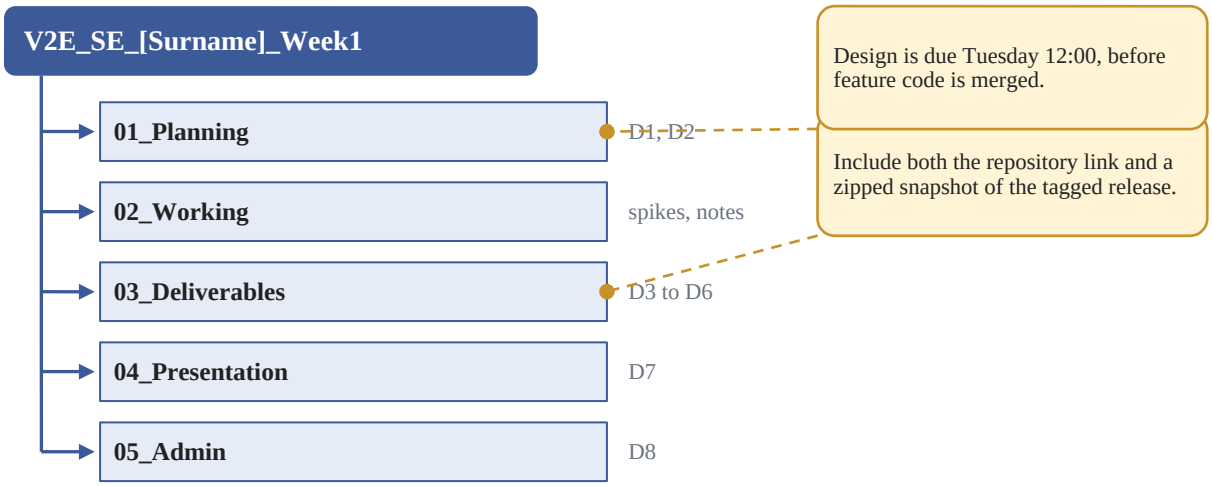


Figure 8.3. Software Engineering submission folder with deliverable placement

8.8 Assessment rubric

Criterion	Weight	Exceeds expectations	Meets expectations	Not yet meeting
Technical quality	30%	Clean structure, correct lifecycle logic, idempotent events, strong tests and secure defaults	Working features with reasonable structure and tests	Features broken, logic incorrect or no meaningful tests

Criterion	Weight	Exceeds expectations	Meets expectations	Not yet meeting
Problem solving	20%	Contradictions resolved, sensible scope negotiation, well reasoned ADRs	Most contradictions found; decisions recorded	Requirements followed blindly or ignored
Communication	20%	Excellent README, guide and demo that a client and a developer can both follow	Clear documentation and demo	Documentation missing or unusable
Delivery	15%	All Must stories done, tagged release, all deliverables on time	Most Must stories done and deliverables on time	Core stories missing or late
Professionalism	15%	Exemplary commit history, pull requests, naming and timesheet	Regular commits and complete admin	Single large commit; admin missing

8.9 Stretch goals

- Containerise the service with Docker and a compose file that starts the service and a PostgreSQL database together.
- Add a simple dashboard for depot staff showing counts of parcels by status and failed attempts today.
- Publish status change events to a message queue abstraction so that notifications can be added later without changing the tracking module.
- Add load testing with a tool such as k6 or Locust and compare results with your non functional requirements.

VET2EMPLOY | TRACK 3 OF 4

Cybersecurity

3

Help a small financial cooperative understand an email account compromise, confirm it is contained, and build a realistic plan to protect its members.

CLIENT ENGAGEMENT (FICTITIOUS)

Ashbourne Valley Credit Union

27	12	9	5
planned hours (minimum 20)	sequential activities	required deliverables	working days Monday to Friday

Note from your supervisor
This is defensive work. You assess evidence and advise. You never test live systems.

Your mission this week

Work from supplied evidence to build a clear timeline of an email account compromise, confirm that containment and recovery are complete, assess the organisation against recognised UK security guidance, and deliver an incident report, a risk register, a prioritised improvement roadmap, updated procedures and a plain English board briefing.

9. Track 3: Cybersecurity

9.1 Welcome note from your Track Supervisor

Welcome to the Cybersecurity track. This week you are joining a Phexara security advisory team that has been asked to help Ashbourne Valley Credit Union after a difficult few weeks. A member of staff was tricked by a convincing email, an attacker gained access to a single mailbox, and for a short period they were able to read and send messages before anyone noticed. The credit union acted, closed the access and now wants an independent view of what happened, whether they are safe, and what they should do next.

Your work this week is entirely defensive and advisory. You are an investigator working from evidence that has already been collected, an assessor comparing the organisation against recognised UK guidance, and an adviser who can explain all of this to a board of volunteer directors who are not technical. You will not touch any live system, and you will not use any offensive tool. Everything you need is in the evidence pack.

Two things matter more to me than anything else. Your conclusions must be supported by evidence that someone else could check, and your advice must be realistic for a small organisation with a small budget. A beautiful plan that Ashbourne Valley cannot afford or deliver is of no use to them. Read the whole brief, respect the boundaries, and remember that behind every log line there are real members whose savings depend on this cooperative being trustworthy.

Your Track Supervisor, Cybersecurity Track
Phexara Security Advisory Practice

9.2 The client: Ashbourne Valley Credit Union

Fact	Detail
Organisation	A member owned financial cooperative founded in 1987, authorised and regulated in the UK, offering savings accounts and affordable loans
Footprint	Head office in Ashbourne, Derbyshire, with branches in Leek and Uttoxeter; about 18,000 members
People	46 staff and volunteers and a board of nine volunteer directors
Technology	Microsoft 365 for email and files, a hosted core banking platform from a specialist supplier, about 55 Windows laptops and desktops, a small on premises file server and IT support from a local managed service provider
Your sponsor (persona)	Helen Matthews, Chief Executive. Calm, decisive and concerned about member trust and the regulator
Other stakeholders	Raj Patel, Finance and Operations Manager; Chris Doyle, account manager at the managed service provider; Margaret Ellis, Chair of the Audit and Risk Committee

9.3 The problem you are solving

What the organisation has told us

In early September 2026 a finance officer received an email that appeared to come from a trusted service, asking her to view a shared document. She followed the link to a web page that imitated a familiar sign in screen and entered her work credentials. The page showed an error and she continued with her day. Over the following few days an attacker used those credentials to sign in to her mailbox from an unfamiliar location, created a hidden rule that moved certain incoming messages out of sight, and read business correspondence.

Using what they had read, the attacker sent a message from the mailbox asking a colleague to change the bank details held for a regular supplier before an invoice was paid. The change was made and a payment was released. The credit union's own bank flagged the new account as suspicious and recalled the payment the same day, so no money was ultimately lost. The incident was formally recognised when the genuine supplier asked why they had not been paid.

The managed service provider responded by resetting the affected password, removing the hidden rule and switching on multifactor authentication for that account. The board is grateful but uneasy. They do not know whether the response was complete, whether anything else was touched, or how exposed they remain. That is why Phexara has been engaged.

How we understand the incident so far

The diagram below sets out the sequence of events as currently understood from the organisation's own account. Part of your job in Activity 3 is to confirm, correct or extend this picture using the evidence pack, and to turn each stage into a question that the evidence can answer.

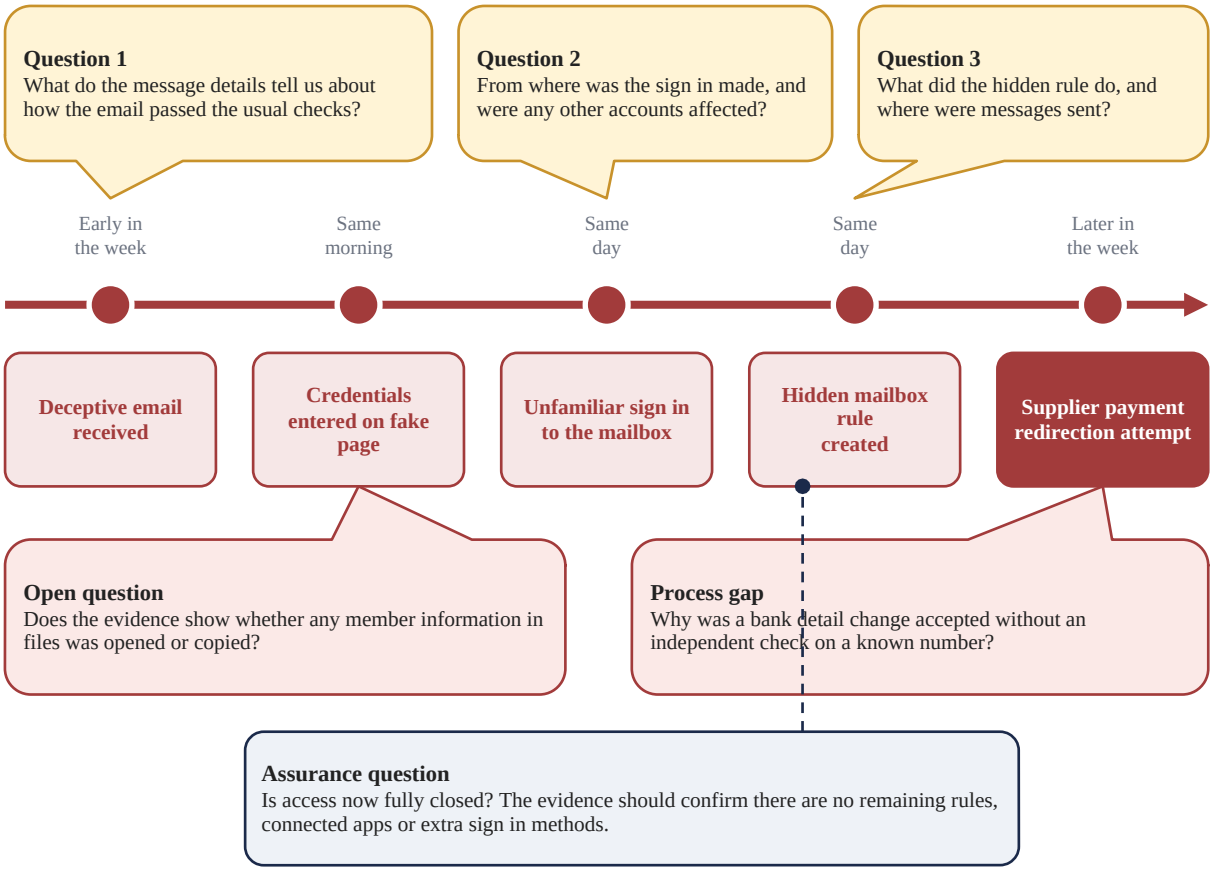


Figure 9.1. Current understanding of the incident timeline, with investigation questions annotated

The wider picture

During early conversations and a review of the managed service provider's documentation, Phexara noted several features of the environment that may have made the incident more likely or that could matter next time. The simplified diagram below summarises them. You will test these observations against the evidence and the organisation's answers to your questions, and decide which represent real risk.

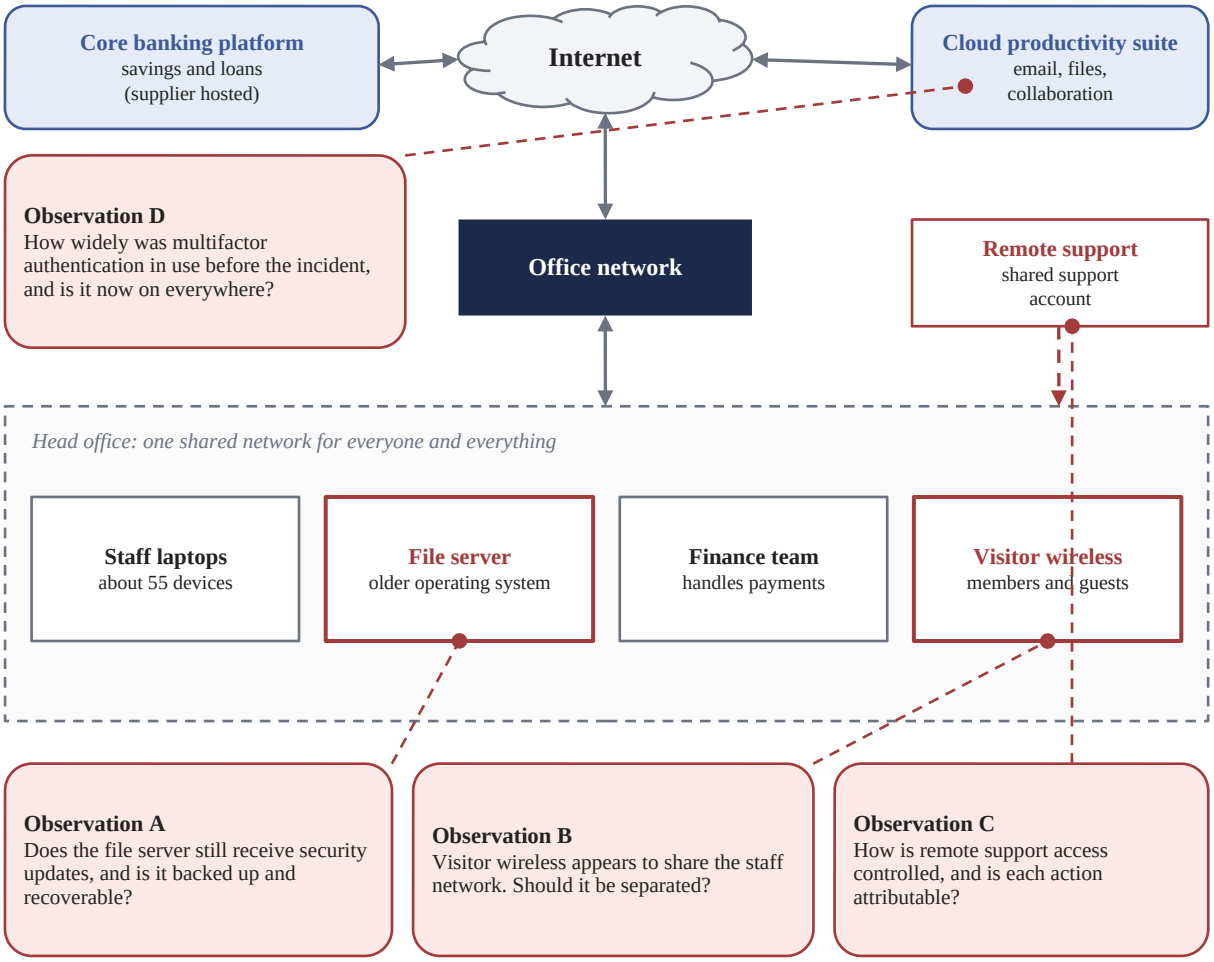


Figure 9.2. Simplified view of Ashbourne Valley systems, with observations to verify annotated

What the board needs from you

1. A clear, evidence based account of what happened, when it happened and what was affected, written so that a nontechnical director can follow it.
2. Independent assurance that access has been fully closed and recovery is complete, or a precise list of the actions still needed to reach that point.
3. An honest assessment of the organisation's security against the five technical controls of the UK Government's Cyber Essentials scheme and the National Cyber Security Centre's guidance for small organisations.
4. A risk register that the Audit and Risk Committee can own and review every quarter.
5. A prioritised and costed improvement roadmap for the next twelve months that is realistic for a small cooperative.
6. Updated procedures for approving supplier bank detail changes and for responding to future incidents, plus a short staff awareness briefing.
7. A factual summary that helps the board decide whether the incident needs to be reported to its regulators or to the Information Commissioner's Office. You set out the facts; you do not give legal advice.

Boundaries of the engagement

These boundaries exist to keep your work lawful, safe and genuinely useful. Staying inside them is assessed, and in real life it is the difference between a trusted professional and a liability.

Area	What you may do	What you must not do
Live systems	Work only from the supplied evidence pack and the organisation's written answers	Sign in to, scan, probe or test any real Ashbourne Valley system or account
Tools	Use ordinary analysis tools such as spreadsheets, a text editor and diagramming software to review evidence	Use any offensive security or exploitation tool against anyone
Personal data	Treat every name, email, account number and member record in the evidence as real personal data; mask it in your outputs	Copy personal data into external services or keep it beyond the cohort
Advice	Set out facts, risks and options clearly	Present yourself as giving regulatory or legal advice

Why this matters

Testing any system without clear written permission can be a criminal offence under the Computer Misuse Act 1990. In this simulation there is never a reason to go near a live system. If you ever feel you need to, stop and ask your supervisor first.

9.4 Your working materials

The evidence pack below has been collected for you, as it would be by an incident handler before an adviser is brought in. Read the guide first. The evidence has been prepared so that a careful reader can build a complete and defensible timeline from it.

Material	Description	How to use it
avcu_evidence_guide.pdf	An index of the evidence pack and how each item was collected	Read first; it explains what each file is and is not
avcu_signin_history.csv	Account sign in records for the affected mailbox and a sample of other accounts, with times, locations and outcomes	Build the access timeline and spot unfamiliar sign ins
avcu_mailbox_audit.csv	A log of mailbox actions such as rule creation, message access and items sent	Establish what the attacker did inside the mailbox
avcu_message_headers.txt	The technical details of the deceptive email and the fraudulent reply	Understand how the email reached the inbox and how the reply was sent
avcu_rule_details.txt	The definition of the hidden mailbox rule that was found and removed	Describe exactly what the rule did
avcu_msp_actions.pdf	A note from the managed service provider listing the response actions and their times	Assess whether containment and recovery were complete
avcu_asset_and_control_summary.xlsx	A self assessment of devices, accounts, backups and current security settings	Assess posture against Cyber Essentials and NCSC guidance
avcu_payment_process_notes.pdf	A description of how supplier payments and bank detail changes are approved today	Identify the process weakness and design a better control

Supervisor hint

The evidence pack contains at least two details that do not match the organisation's informal account of events. Finding these, and resolving them through the client question window rather than guessing, is part of the assessment.

9.5 Your week at a glance

No.	Activity	When	Effort
1	Induction, brief review and evidence familiarisation	Mon morning	1.5 hours
2	Scoping, questions and investigation plan	Mon afternoon	2.0 hours
3	Timeline reconstruction from evidence	Mon and Tue	3.0 hours
4	Impact and data exposure assessment	Tue	2.5 hours
5	Containment and recovery assurance	Tue and Wed	2.5 hours
6	Posture assessment against UK guidance	Wed	3.0 hours
7	Risk register	Wed and Thu	2.5 hours
8	Improvement roadmap	Thu	2.5 hours
9	Payment control and incident response procedures	Thu	2.0 hours
10	Staff awareness briefing	Fri morning	1.5 hours
11	Incident report and board briefing	Fri morning	2.5 hours
12	Showcase, handover and reflection	Fri afternoon	1.5 hours
	Total planned effort	Monday to Friday	27.0 hours

9.6 Sequential activities

Complete the activities in order. Your findings about what happened (Activities 3 to 5) must come before your judgements about risk and improvement (Activities 6 to 9), which in turn feed the report and board briefing at the end of the week.

Activity 1: Induction, brief review and evidence familiarisation

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Monday morning	**1.5 hours**	**None**	**Evidence inventory**

Objective

Understand the engagement, confirm you can open every item of evidence, and set up a disciplined way of working that protects the integrity of that evidence.

What you must do, in order

1. Attend the induction call and record any clarifications.
2. Read this entire track brief, including the boundaries of the engagement and the deliverables.
3. Read the evidence guide and confirm that every listed item is present and opens.

4. Create your submission folder and copy the evidence into a read only location. Never edit the original evidence.
5. Create an evidence inventory that lists each item, what it is, how it was collected and any limitations.
6. Start your timesheet and reflection log.

You are done when

- Every evidence item is listed in your inventory.
- Original evidence is stored read only.
- Your timesheet and reflection log are started.

Activity 2: Scoping, questions and investigation plan

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Monday afternoon	**2.0 hours**	**Activity 1**	**D1 Investigation Plan**

Objective

Turn the board's needs into a precise set of questions and a plan your supervisor can approve.

What you must do, in order

1. Write down the questions the investigation must answer, grouped under what happened, what was affected, and are we now safe.
2. For each question, note which evidence item you expect to answer it and what a convincing answer would look like.
3. List the assumptions you are making until the client replies, and the limitations of the evidence.
4. Write between five and eight clarification questions for the client persona, prioritising those that would change your conclusions. Submit them before the question window closes.
5. Draft a one page to two page investigation plan covering questions, approach, assumptions, a day by day schedule and how you will keep evidence and personal data safe.
6. Send the plan to your supervisor for approval.

You are done when

- The plan maps every board need to evidence and method.
- At least five clarification questions have been submitted.
- Your supervisor has acknowledged the plan.

> **A strong investigation question**
>
> Weak: Was the email bad? Strong: Do the message details show whether the deceptive email failed the standard sender checks, and would the organisation's current settings have quarantined it?

Activity 3: Timeline reconstruction from evidence

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Monday and Tuesday	**3.0 hours**	**Activity 2**	**D2 Incident Timeline**

Objective

Build a precise, evidence backed timeline of the incident from first contact to response.

What you must do, in order

1. Work through the sign in history and mailbox audit and record every relevant event with its date, time and time zone, what happened and the evidence reference.
2. Use the message headers to describe how the deceptive email arrived and how the fraudulent reply was sent, in plain language.
3. Describe exactly what the hidden rule did, using the rule details.
4. Convert all times to a single time zone and note where the evidence is uncertain or where times conflict.
5. Compare your timeline with the organisation's informal account and highlight the points where they differ.
6. Present the timeline both as a table and as a simple annotated diagram.

You are done when

- Every event in the timeline cites an evidence reference.
- Times are normalised to one time zone with uncertainties noted.
- Differences from the informal account are clearly flagged.

Activity 4: Impact and data exposure assessment

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Tuesday	**2.5 hours**	**Activity 3**	**D3 Impact Assessment**

Objective

Decide, on the evidence, what was actually affected, so that the board can judge seriousness and reporting obligations.

What you must do, in order

1. Establish which mailbox and which accounts were accessed, and for how long.
2. Determine from the evidence whether any files or member information were opened, downloaded or sent, and state clearly where the evidence is silent.
3. Describe the attempted financial impact and its outcome.
4. Assess the potential impact on members, on the organisation's finances and on its reputation, separating what is known from what is possible but unproven.
5. Summarise the facts that are relevant to whether the incident should be reported to a regulator or to the Information Commissioner's Office, without offering legal advice.
6. Mask all personal data in your output, using references rather than real names.

You are done when

- Known impact and possible but unproven impact are clearly separated.
- The data exposure position is stated honestly, including gaps in the evidence.
- No unmasked personal data appears in the assessment.

Activity 5: Containment and recovery assurance

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Tuesday and Wednesday	**2.5 hours**	**Activity 4**	**D4 Assurance Checklist**

Objective

Give the board an independent view on whether access is truly closed and recovery is complete.

What you must do, in order

1. List the actions the managed service provider took and the time of each, from their note.
2. Build a containment and recovery checklist covering the things that should be verified after this kind of incident, such as credentials reset, extra sign in methods removed, hidden rules and message forwarding removed, connected applications reviewed and other accounts checked for the same pattern.
3. Mark each item as evidenced as complete, not yet evidenced or not done, citing the evidence.
4. Identify any gap that would allow access to continue, and recommend a specific action for each.
5. State a clear overall conclusion: is the organisation contained and recovered, or not yet, and what remains.

You are done when

- Every checklist item has a status backed by evidence.
- Each gap has a specific recommended action.
- There is a clear overall assurance conclusion.

Assessing posture against UK guidance

In Activity 6 you will assess Ashbourne Valley against the five technical controls of Cyber Essentials. The diagram below reminds you of those controls and shows how each one relates to this incident. Use it to keep your assessment anchored to recognised guidance rather than personal opinion.

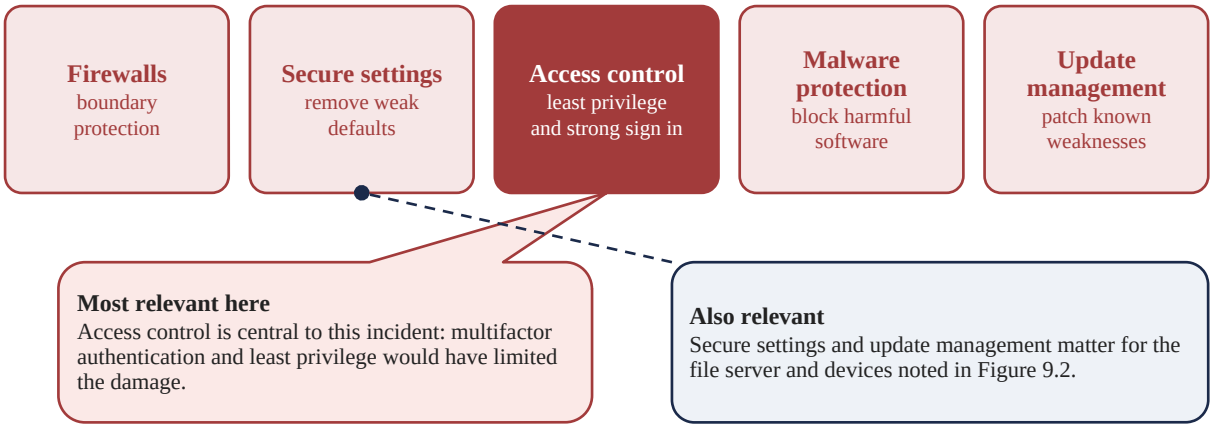


Figure 9.3. The five Cyber Essentials technical controls, annotated with their relevance to this incident

Activity 6: Posture assessment against UK guidance

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Wednesday	**3.0 hours**	**Activity 5**	**D5 Posture Assessment**

Objective

Assess the organisation against recognised UK guidance so that your advice is grounded and defensible.

What you must do, in order

1. Using the asset and control summary, assess Ashbourne Valley against each of the five Cyber Essentials technical controls. For each, record the current position, the evidence and a rating of met, partly met or not met.
2. Compare the organisation's practices with the National Cyber Security Centre's Small Business Guide and Small Charity Guide, noting the most important gaps.
3. Pay particular attention to the themes this incident exposed: resistance to deceptive emails, strong sign in, the supplier payment process, and the ability to detect and respond.
4. Summarise strengths as well as weaknesses. A fair assessment builds trust.
5. Produce a simple one page heat view of the controls so that the board can see the picture at a glance.

You are done when

- Every Cyber Essentials control has a rating and supporting evidence.
- The most important gaps are linked to the incident.
- The assessment is balanced and includes strengths.

Activity 7: Risk register

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Wednesday and Thursday	**2.5 hours**	**Activity 6**	**D6 Risk Register**

Objective

Capture the risks in a form the Audit and Risk Committee can own and review every quarter.

What you must do, in order

1. Create a risk register with, for each risk, an identifier, a description in cause and effect form, the affected assets, the current likelihood and impact, an overall rating, the existing controls, the proposed further action, an owner and a target date.
2. Include at least ten risks drawn from the incident and the posture assessment, such as deceptive emails, weak sign in, the payment change process, the older file server, flat networking and limited monitoring.
3. Use a consistent and documented scale for likelihood and impact.
4. Show both the current rating and the expected rating after the proposed actions.
5. Keep the language plain so that a volunteer director can understand each risk.

You are done when

- At least ten risks are recorded in the required format.
- The scoring scale is documented and applied consistently.
- Each risk has an owner and a target date.

Activity 8: Improvement roadmap

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Thursday	**2.5 hours**	**Activity 7**	**D7 Improvement Roadmap**

Objective

Turn the risks into a realistic, prioritised and costed plan a small organisation can deliver.

What you must do, in order

1. Group the recommended actions into immediate (within two weeks), short term (within three months) and medium term (within twelve months).
2. For each action, estimate the effort, a rough cost band, the responsible party and the risks it reduces.
3. Prioritise using the balance of risk reduction against cost and effort, favouring high impact and low cost actions first, such as enabling strong sign in everywhere and improving the payment change process.
4. Show the roadmap as a simple timeline that the board can follow.
5. Be honest about what the organisation can realistically achieve with its resources, and note where outside help may be needed.

You are done when

- Actions are grouped by timeframe and prioritised by value.
- Each action has an owner, an effort estimate and a cost band.
- The plan is realistic for a small cooperative.

Activity 9: Payment control and incident response procedures

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Thursday	**2.0 hours**	**Activity 8**	**D8 Procedures**

Objective

Fix the process that let the fraud happen and give the organisation a simple plan for next time.

What you must do, in order

1. Design an improved procedure for approving changes to supplier bank details, including an independent check using contact details already held, a second approver and a record of each change. Show it as a clear flow.
2. Write a short, practical incident response procedure suited to a small organisation: how to recognise an incident, who to tell, the first steps to take, how to preserve evidence and when to escalate to the managed service provider and the board.
3. Include a simple one page incident contact and escalation card.
4. Keep both procedures short enough that busy staff will actually use them.

You are done when

- The payment change procedure removes the single point of failure that was exploited.
- The incident response procedure is realistic and fits on a few pages.
- A one page escalation card is included.

Activity 10: Staff awareness briefing

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Friday morning	**1.5 hours**	**Activity 9**	**D9 part one, Awareness Briefing**

Objective

Help the people who are the first line of defence to recognise and resist the next attempt.

What you must do, in order

1. Prepare a short awareness briefing for all staff that explains, without blame, how this kind of incident happens and how to spot the warning signs.
2. Give practical, memorable advice: how to check an unexpected request, how to treat links and sign in prompts with caution, and how to report a concern quickly.
3. Keep the tone supportive. People who fear blame hide mistakes, which makes the next incident worse.
4. Make it suitable for a fifteen minute team session, with a one page handout.

You are done when

- The briefing explains the risk in plain, blame free language.
- The advice is practical and memorable.
- A one page handout accompanies it.

Activity 11: Incident report and board briefing

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Friday morning	**2.5 hours**	**Activity 10**	**D9 part two, Incident Report and Board Briefing**

Objective

Bring everything together into a clear report and a briefing the board can act on.

What you must do, in order

1. Write an incident report with an executive summary, the timeline, the impact, the assurance position, the posture assessment, the risks and the recommendations, plus appendices for evidence references and assumptions.
2. Write the executive summary so that a nontechnical director understands the whole picture in a single page.
3. Prepare a board briefing of no more than ten slides that tells the story as situation, impact, assurance, risks and plan, and ends with the decisions the board needs to make.
4. Record a version of the briefing of no more than five minutes.
5. Check every claim against your evidence and make sure no unmasked personal data appears anywhere.

You are done when

- The report is complete and every claim is evidenced.
- The executive summary is understandable by a nontechnical reader.
- The briefing ends with clear decisions for the board.

Activity 12: Showcase, handover and reflection

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Friday afternoon	**1.5 hours**	**Activity 11**	**D9 part three and D10 Reflection**

Objective

Present your findings as you would to the board and hand over cleanly.

What you must do, in order

1. Deliver your board briefing to the cohort and supervisors as if they were the board.
2. Answer questions, including the difficult ones about what remains uncertain.
3. Hand over your risk register and roadmap in a form the Audit and Risk Committee can maintain.

4. Complete your timesheet, reflection log and declaration of any AI tool use.
5. Check names and upload every deliverable by 17:00.

You are done when

- The briefing is delivered within time.
- The handover materials are usable by the committee.
- Timesheet and reflection log are complete.

9.7 Required deliverables

Ref	Deliverable	Format and file name	Due	Acceptance criteria
D1	Investigation Plan	PDF, V2E_CY_[Surname]_D1_InvestigationPlan_v1.pdf	Mon 17:00	Questions mapped to evidence, assumptions, schedule and handling rules
D2	Incident Timeline	PDF or XLSX plus a diagram	Tue 17:00	Every event evidenced, times normalised, conflicts flagged
D3	Impact Assessment	PDF, V2E_CY_[Surname]_D3_Impact_v1.pdf	Wed 12:00	Known and possible impact separated; personal data masked
D4	Containment and Recovery Assurance	XLSX or PDF checklist	Wed 17:00	Each item evidenced; gaps and actions listed; clear conclusion
D5	Posture Assessment	PDF with a one page heat view	Thu 12:00	Five controls rated with evidence; balanced; linked to the incident
D6	Risk Register	XLSX, V2E_CY_[Surname]_D6_RiskRegister_v1.xlsx	Thu 17:00	Ten or more risks with consistent scoring, owners and target dates
D7	Improvement Roadmap	PDF with a timeline	Fri 17:00	Prioritised, costed and realistic for a small organisation
D8	Procedures	PDF with flow diagrams	Fri 17:00	Improved payment change control and a practical incident response procedure
D9	Awareness Briefing, Incident Report and Board Briefing	PDF, PPTX and video link	Fri 17:00	Report evidenced; summary nontechnical; briefing ends with decisions
D10	Timesheet and Reflection Log	XLSX and PDF	Fri 17:00	At least twenty hours logged; AI use declared; decision log complete

Where each deliverable goes

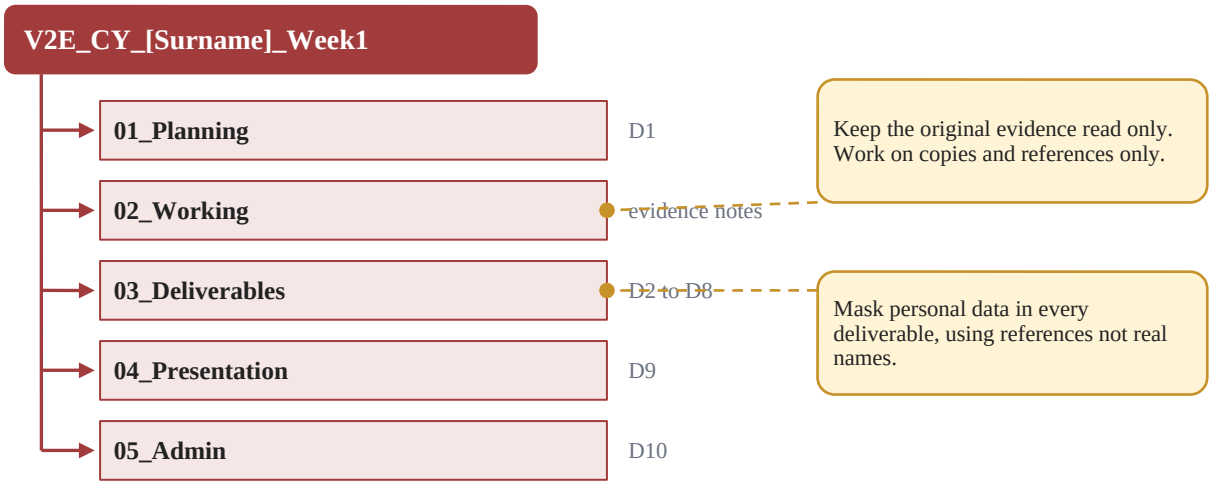


Figure 9.4. Cybersecurity submission folder with deliverable placement

9.8 Assessment rubric

Criterion	Weight	Exceeds expectations	Meets expectations	Not yet meeting
Technical quality	30%	Rigorous, fully evidenced timeline, impact and assurance; posture assessment grounded in recognised guidance	Sound findings mostly backed by evidence	Claims not supported by evidence or guidance ignored
Problem solving	20%	Evidence conflicts found and resolved; honest about uncertainty; sensible risk judgements	Most issues found and documented	Conflicts missed; uncertainty hidden
Communication	20%	Report and briefing a nontechnical board can act on; clear visuals and plain language	Clear report and briefing	Jargon heavy or unclear; board could not act
Delivery	15%	All deliverables complete, on time and polished	All deliverables complete and on time	Missing or late deliverables
Professionalism	15%	Exemplary evidence handling, data masking, naming and timesheet; strict respect for boundaries	Boundaries respected; admin complete	Boundaries or data handling breached; admin missing

9.9 Stretch goals

- Design a simple, low cost monitoring and alerting approach that would have detected this incident sooner, using tools the organisation already has.
- Draft a one page template for reporting an incident to the relevant authorities, with the facts a small organisation would need to supply.
- Create a short tabletop exercise the board could run once a year to practise responding to a similar incident.

- Propose a simple set of security measures the organisation could ask its managed service provider to commit to in its contract.

VET2EMPLOY | TRACK 4 OF 4

Data Engineering

4

Build a reliable, tested data pipeline that turns raw charger telemetry, payments and site data into numbers the business and the regulator can trust.

CLIENT ENGAGEMENT (FICTITIOUS)

Voltaway Charging Networks Ltd

26	12	8	5
planned hours (minimum 20)	sequential activities	required deliverables	working days Monday to Friday

Note from your supervisor
Rerunning your pipeline twice must give exactly the same answer. Idempotency is assessed.

Your mission this week

Design and build a layered batch pipeline that ingests four messy sources, cleans and conforms them, tests their quality automatically, models them into a star schema and publishes reconciled revenue and charger reliability marts, with orchestration, monitoring and documentation that Voltaway's small team can run without you.

10. Track 4: Data Engineering

10.1 Welcome note from your Track Supervisor

Welcome to the Data Engineering track. This week you are joining a Phexara data platform squad that has been engaged by Voltaway Charging Networks, a fast growing operator of public electric vehicle charge points. Voltaway's drivers can charge their cars reliably, most of the time. Voltaway's managers, however, cannot answer simple questions about their own business with confidence. How much money did we take last month? Which chargers keep failing? Are we meeting our reliability obligations? Every answer depends on who ran the query and when.

Analysts and data scientists get the credit when a chart impresses a board, but none of it is possible without the plumbing you are about to build. Your work this week is to make data arrive on time, in the right shape, with the right meaning and with evidence that it is correct. That means thinking carefully about raw versus cleaned data, about what happens when a file arrives twice or late, about schemas that change without warning and about tests that fail loudly before a wrong number reaches the finance team.

Keep your design simple, make every step repeatable and write down every rule you apply. I would rather see a small pipeline that is idempotent, tested and documented than a sprawling one that only works on your laptop on a good day. Talk to me early if a source behaves strangely. Some of them have been built to.

Your Track Supervisor, Data Engineering Track
Phexara Data Platform Practice

10.2 The client: Voltaway Charging Networks Ltd

Fact	Detail
Business	Operator of public electric vehicle charge points, founded in 2019 and based in Milton Keynes
Network	About 620 charge points across 140 sites in England and Wales, a mix of 7 kW destination chargers, 50 kW rapid chargers and 150 kW ultra rapid chargers
Activity	Roughly 95,000 charging sessions a month, paid by contactless card, the Voltaway app or roaming partner accounts
People	About 120 staff, including a two person data team and a field engineering team
Your sponsor (persona)	Nadia Okonkwo, Head of Data. Technically strong, short of time and keen to replace fragile scripts with something the team can trust
Other stakeholders	Marcus Bell, Chief Financial Officer; Ellie Shaw, Head of Field Operations; Tom Varga, Compliance Manager

10.3 The problem you are solving

Background

Voltaway doubled the size of its network in two years. Data arrives from several systems that were chosen quickly and never designed to work together. The charge points report session and status data to a charge point management system using the industry standard Open Charge Point Protocol (OCPP). Card payments are settled by a payment processor that sends a daily settlement file. Site details live in a spreadsheet maintained by the property team. Tariffs are held in a table that the commercial team edits by hand.

Today, one analyst runs a collection of scripts every Monday that copy data from these sources into spreadsheets and a reporting database. The scripts break whenever a file format changes, they cannot be rerun safely, and nobody else understands them. When the analyst was on holiday in August, no reports were produced for two weeks.

What is going wrong

- **Revenue does not reconcile.** Revenue calculated from charging sessions differs from cash received from the payment processor by about 3.2 percent, and nobody can explain the gap.
- **Reliability figures are disputed.** Field Operations calculates charger availability from fault tickets; the charge point management system shows different figures. Voltaway is required to monitor and report reliability for its rapid network, and the Compliance Manager cannot defend either number.
- **Duplicates and gaps are common.** Charger messages are sometimes sent twice, and some chargers lose connectivity and upload a backlog of sessions hours or days later.
- **Schemas change silently.** A firmware update in July renamed two fields in the session data, which caused a month of energy figures to be recorded as zero.
- **There is no history.** Site and tariff changes overwrite the previous values, so past sessions are reported at today's price and against today's site details.

What the client has asked for

Nadia Okonkwo wants a first version of a proper analytical data platform that the data team can run and extend. It must load the sources into a central store, keep the raw data untouched, clean and standardise it, test it automatically, model it for analysis and publish two trusted outputs: a monthly revenue reconciliation for the CFO and a daily charger reliability mart for Field Operations and Compliance.

Target pipeline architecture

The diagram below shows the layered architecture that Phexara has recommended. It follows the widely used bronze, silver and gold pattern. You may choose different tools, but you must keep the layers and explain any departure in an architecture decision record.

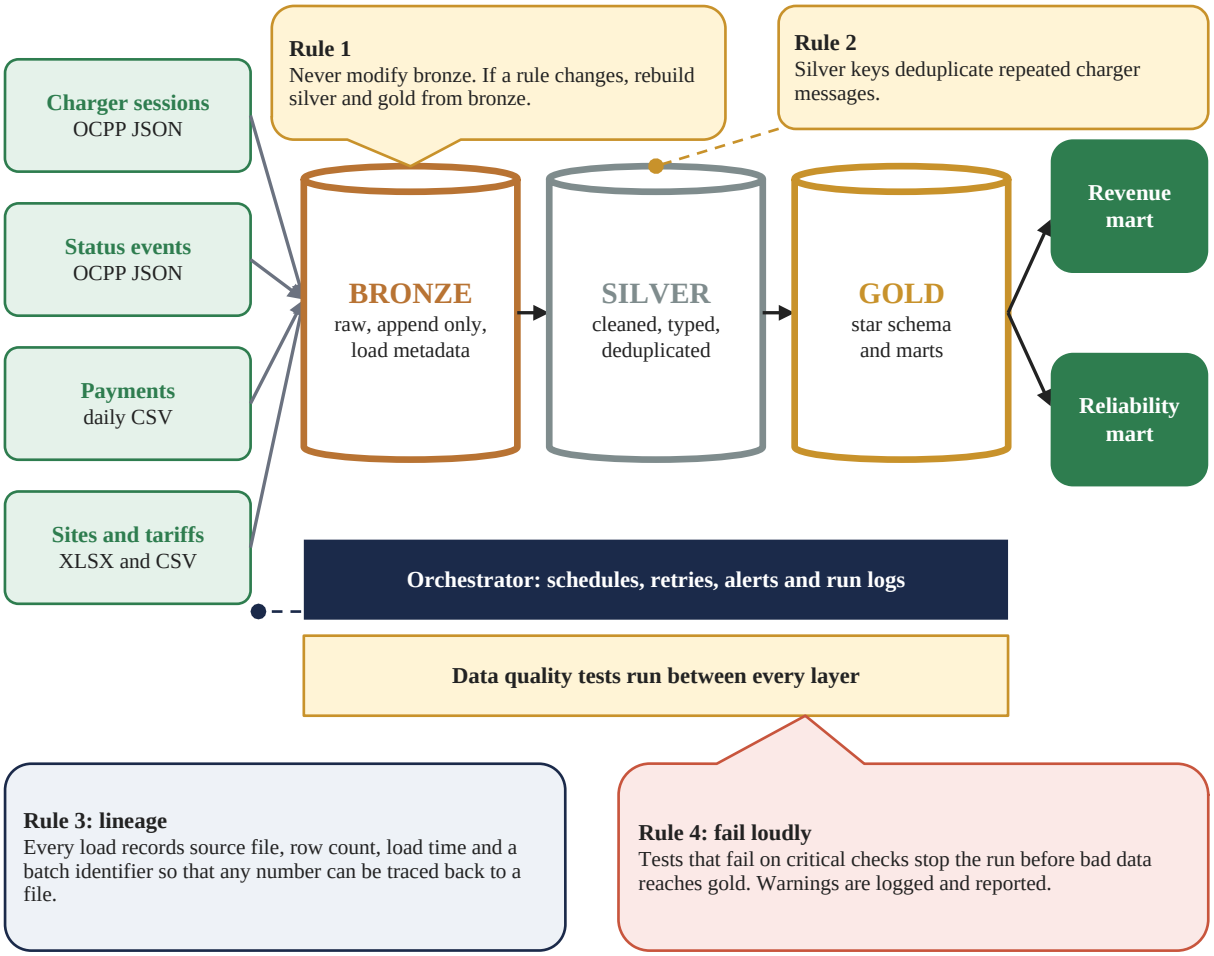


Figure 10.1. Target layered pipeline architecture, annotated with design rules

Scope and constraints

In scope	Out of scope
Batch ingestion of the four sources for six months of history	Real time streaming ingestion
Bronze, silver and gold layers with tests	Machine learning or forecasting
Star schema plus revenue and reliability marts	Building dashboards beyond a simple validation view
Orchestration, logging, documentation and handover	Cloud production deployment and cost optimisation

- Use Python and SQL. A local analytical database such as DuckDB or PostgreSQL is recommended. Spark is permitted but not required for this data volume.
- Transformations may be written in SQL with a framework such as dbt, or in Python with pandas or Polars. Data quality tests may use dbt tests, Great Expectations, Soda or plain assertions.
- Orchestration may use Apache Airflow, Prefect, Dagster or a well structured command line runner with a scheduler. Justify your choice for a two person team.
- Everything must run from a clean checkout using documented commands. Use Git throughout.

- Card data in the payments file is already tokenised. Do not attempt to reverse tokens, and do not store driver email addresses in gold.

10.4 Your working materials

File	Grain and size	Key fields	Known or suspected issues
sessions/*.json	One file per day, one record per charging session; about 570,000 sessions	session_id, charger_id, connector_id, start_ts, stop_ts, energy_kwh, payment_method, auth_token	Duplicates, late arriving files, fields renamed in July, timestamps in mixed time zones
status/*.json	One record per status change; about 2.1 million events	charger_id, connector_id, status, error_code, event_ts	Out of order events, gaps when chargers are offline
payments/ settlement_*.csv	One file per day, one row per transaction; about 480,000 rows	txn_ref, settled_ts, amount_gbp, fee_gbp, card_token, merchant_ref	Refunds as separate rows, currency stored as text with a pound sign, one file delivered twice
sites.xlsx	One row per site; 140 rows	site_id, site_name, postcode, region, host_type, opened_date	Edited by hand, two sites with duplicate IDs
chargers.csv	One row per charger; about 620 rows	charger_id, site_id, power_kw, charger_class, commissioned_date	Decommissioned chargers still listed
tariffs.csv	One row per tariff change; about 90 rows	site_id, charger_class, price_per_kwh, valid_from	No end dates; overlapping validity
data_contract_notes .pdf	Reference	Field meanings and business rules from Nadia's team	Incomplete; reliability definition not agreed

Supervisor hint

On Wednesday at 10:00 a new sessions file will be added to the materials folder covering a day that you have already loaded. It contains corrected records. Your pipeline must handle it without creating duplicates or losing the corrections.

10.5 Your week at a glance

No.	Activity	When	Effort
1	Induction, brief review and environment set up	Mon morning	1.0 hours
2	Source discovery and data contract	Mon	2.0 hours
3	Pipeline architecture and design decisions	Mon and Tue	2.5 hours
4	Repository, project structure and local warehouse	Tue morning	1.5 hours
5	Bronze ingestion with load metadata	Tue	3.0 hours
6	Silver cleaning, typing and deduplication	Wed	3.5 hours
7	Automated data quality tests	Wed	2.5 hours
8	Gold dimensional model	Thu	3.0 hours
9	Revenue and reliability marts	Thu	2.0 hours
10	Orchestration, scheduling and monitoring	Thu and Fri	2.0 hours

No.	Activity	When	Effort
11	Documentation, lineage and runbook	Fri morning	1.5 hours
12	Demonstration, handover and reflection	Fri afternoon	1.5 hours
	Total planned effort	Monday to Friday	26.0 hours

10.6 Sequential activities

Complete the activities in order. Each layer depends on the one before it, and tests must exist before you build gold models on top of silver.

Activity 1: Induction, brief review and environment set up

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Monday morning	**1.0 hours**	**None**	**Working environment**

Objective

Understand the engagement and confirm that your tools work before you start designing.

What you must do, in order

1. Attend the induction call and record clarifications.
2. Read this entire track brief and the data contract notes.
3. Install Python, Git, your chosen database and any framework you plan to use, and confirm that you can create a table and query it.
4. Copy the materials into a read only landing folder that will act as your simulated source system.
5. Start your timesheet and reflection log.

You are done when

- Your toolchain works end to end on a trivial example.
- Source files are stored read only.

Activity 2: Source discovery and data contract

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Monday	**2.0 hours**	**Activity 1**	**D1 Source Profile and Data Contract**

Objective

Understand every source in detail and agree what each field means before you move any data.

What you must do, in order

1. Profile each source: record counts, fields, types, null rates, distinct counts, date ranges and sample values. Note any file whose structure differs from the others.
2. Identify the natural key of each source and test whether it is truly unique.
3. Find the schema change in the session files and document the old and new field names and the date of the change.

4. Write a data contract for each source: owner, delivery frequency, expected arrival time, schema, key, allowed values, freshness expectation and what should happen when the contract is broken.
5. Write clarification questions for the client, including how reliability should be defined and how refunds should be treated in revenue. Submit them before the window closes.

You are done when

- Every source has a profile and a written contract.
- The schema change is documented with evidence.
- Clarification questions are submitted.

> **A useful reliability question**
> Should a charger count as available when it is reporting Available but has failed its last three sessions? The answer changes the reliability figure, so ask rather than assume.

Activity 3: Pipeline architecture and design decisions

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Monday and Tuesday	**2.5 hours**	**Activity 2**	**D2 Design Pack**

Objective

Design the pipeline end to end so that the build is consistent and every choice is explained.

What you must do, in order

1. Draw your pipeline architecture, based on Figure 10.1, showing tools, layers, schedules and data flows.
2. Define your naming conventions for schemas, tables and columns.
3. Decide your loading strategy for each source: full refresh, append or incremental merge, and how late and corrected files are handled.
4. Decide how you will keep history for sites and tariffs, for example as slowly changing dimensions of type 2 with valid from and valid to dates.
5. Write at least three architecture decision records covering tool choice, incremental loading and history handling.
6. Share the design with your supervisor before building beyond bronze.

You are done when

- The design pack contains an architecture diagram, conventions, loading strategies and three ADRs.
- Your supervisor has seen the design.

Activity 4: Repository, project structure and local warehouse

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Tuesday morning	**1.5 hours**	**Activity 3**	**Repository skeleton**

Objective

Set up a clean, conventional project that another engineer would recognise immediately.

What you must do, in order

1. Create a repository named voltaway_pipeline with a README, a .gitignore and a dependency file.
2. Create a folder structure that separates ingestion code, transformations, tests, orchestration and documentation.
3. Create the bronze, silver and gold schemas in your warehouse with a single command.
4. Load configuration such as file paths and connection details from environment variables or a configuration file that is not committed with secrets.
5. Add a linter and a formatter and run them.

You are done when

- The warehouse schemas can be created from a clean checkout.
- No secrets or source data are committed.

Activity 5: Bronze ingestion with load metadata

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Tuesday	**3.0 hours**	**Activity 4**	**Bronze layer loaded**

Objective

Land every source in the warehouse exactly as received, with enough metadata to trace any record back to its file.

What you must do, in order

1. Write ingestion code for each source that reads the files and writes them to bronze tables without changing values. Store JSON either as raw text or as loosely typed columns.
2. Add load metadata to every row: source file name, file checksum, load timestamp and batch identifier.
3. Record each run in a load log table: source, files processed, rows read, rows loaded, start time, end time and status.
4. Make ingestion idempotent at file level: a file with the same checksum must not be loaded twice. Prove this using the payments file that was delivered twice.
5. Run the ingestion for all six months and reconcile row counts in bronze with your source profile.

You are done when

- Bronze row counts match the source profile.
- Running ingestion twice loads nothing new the second time.

- Every bronze row can be traced to a file.

Activity 6: Silver cleaning, typing and deduplication

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Wednesday	**3.5 hours**	**Activity 5**	**Silver layer built**

Objective

Turn raw records into clean, typed, deduplicated and conformed tables that analysts can trust.

What you must do, in order

1. Cast every field to the correct type. Convert all timestamps to UTC and keep the original value for audit.
2. Map the renamed session fields so that records before and after the July change land in the same columns.
3. Deduplicate sessions and status events using their natural keys, keeping the most recent version of a record when a corrected file arrives.
4. Parse the payment amounts to numbers, separate refunds from sales and link payments to sessions where the data allows.
5. Fix or flag the duplicate site identifiers and decommissioned chargers, recording each rule in a transformation log.
6. Load the corrected sessions file released on Wednesday and show that corrections replace the earlier values without creating duplicates.

You are done when

- Silver tables have consistent types and UTC timestamps.
- No duplicate natural keys remain in silver.
- The corrected file is handled correctly and the evidence is recorded.

Activity 7: Automated data quality tests

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Wednesday	**2.5 hours**	**Activity 6**	**D4 Data Quality Report**

Objective

Catch bad data automatically, before it reaches the people who make decisions with it.

What you must do, in order

1. Write tests for uniqueness and not null on every key in silver.
2. Write tests for accepted values (for example status and charger class), referential integrity between sessions, chargers and sites, and valid ranges (for example energy between zero and a sensible maximum for the charger's power and session length).
3. Write freshness tests that warn when a source has not delivered on schedule.
4. Classify each test as critical, which stops the pipeline, or warning, which is logged.

5. Run the tests and record the results in a data quality report, with counts of failing records and the action taken for each.

You are done when

- At least fifteen tests exist and run with one command.
- Critical failures stop the run; warnings are reported.
- The data quality report explains every failure.

Target dimensional model

Use the star schema below as the starting point for your gold layer in Activity 8. The annotations describe requirements that the client has confirmed.

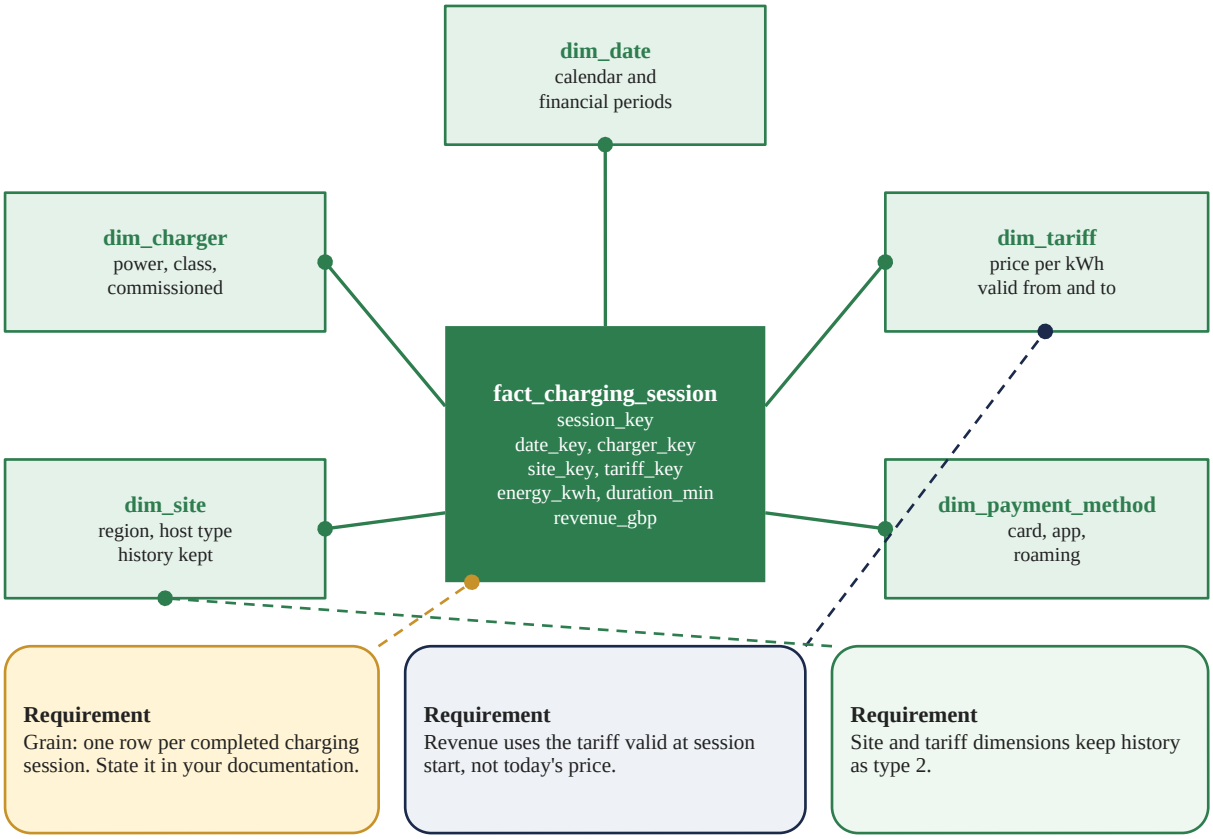


Figure 10.2. Target gold layer star schema for charging sessions, annotated

Activity 8: Gold dimensional model

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Thursday	**3.0 hours**	**Activity 7**	**Gold star schema built**

Objective

Model clean data into a simple star schema that answers business questions quickly and consistently.

What you must do, in order

1. Build the date, charger, site, tariff and payment method dimensions with surrogate keys.

2. Implement history on the site and tariff dimensions using valid from and valid to dates, so that each session joins to the version in force at the time.
3. Build the charging session fact table at the agreed grain and calculate revenue from energy and the correct tariff.
4. Build a charger status fact table at the grain of one row per connector per status interval, with start time, end time and duration.
5. Add tests on the gold layer: row counts that reconcile to silver, no orphan keys and no negative durations.
6. Write a short data dictionary for every gold table and column.

You are done when

- The gold layer passes all tests.
- Sessions join to the tariff version valid at session start.
- A data dictionary covers every gold table.

Activity 9: Revenue and reliability marts

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Thursday	**2.0 hours**	**Activity 8**	**D5 Marts and Reconciliation**

Objective

Publish the two outputs the client actually asked for, and explain the revenue gap.

What you must do, in order

1. Build a monthly revenue mart by site, charger class and payment method, comparing expected revenue from sessions with settled revenue from payments.
2. Investigate the reconciliation gap. Quantify how much is explained by refunds, processor fees, unpaid or failed sessions, roaming settlements, late files and timing differences. State what remains unexplained.
3. Build a daily reliability mart by charger showing minutes in each status and availability percentage, using the definition agreed with the client or your documented assumption.
4. Identify the ten least reliable rapid chargers and the most common error codes.
5. Write a two page reconciliation note for the CFO in plain English.

You are done when

- The revenue gap is broken down into quantified causes.
- The reliability definition is stated and applied consistently.
- The CFO note is understandable without technical knowledge.

Activity 10: Orchestration, scheduling and monitoring

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Thursday and Friday	**2.0 hours**	**Activity 9**	**D3 Orchestrated Pipeline**

Objective

Make the pipeline run itself, in the right order, and tell someone when it goes wrong.

What you must do, in order

1. Define the pipeline as a set of tasks with dependencies: ingest, build silver, test silver, build gold, test gold and build marts.
2. Configure a daily schedule, retries with a delay for transient failures and a timeout for each task.
3. Add alerting for failures, for example a log message, an email stub or a message to a chat webhook stub.
4. Add a backfill command that rebuilds a chosen date range from bronze.
5. Run the whole pipeline twice from a clean warehouse and confirm that the results are identical.

You are done when

- The full pipeline runs with one command or one scheduled trigger.
- Two clean runs produce identical gold tables.
- A failed critical test stops the run and raises an alert.

Activity 11: Documentation, lineage and runbook

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Friday morning	**1.5 hours**	**Activity 10**	**D6 Documentation and Runbook**

Objective

Make the platform understandable and operable by Voltaway's two person data team.

What you must do, in order

1. Update the README with an overview, architecture diagram, set up steps and commands.
2. Produce a lineage view that shows how each gold column is derived from bronze sources. A diagram or a generated documentation site is acceptable.
3. Write a runbook covering daily checks, what to do when each critical test fails, how to load a late file, how to backfill and how to add a new source.
4. List known limitations and a roadmap for the next three months.

You are done when

- A new engineer can run the pipeline from the README alone.
- The runbook covers every critical test failure.

Activity 12: Demonstration, handover and reflection

SCHEDULED	EFFORT	DEPENDS ON	PRODUCES
Friday afternoon	**1.5 hours**	**Activity 11**	**D7 Demo and D8 Reflection**

Objective

Show the client that the platform works and hand it over with confidence.

What you must do, in order

1. Record a demonstration of no more than five minutes: a full run, a failed test stopping the pipeline, the late corrected file being handled and the two marts answering the client's questions.
2. Tag a release in your repository with release notes.
3. Complete your timesheet, reflection log and AI use declaration.
4. Check names and upload every deliverable by 17:00.

You are done when

- The demonstration shows idempotency and failure handling.
- Timesheet and reflection log are complete.

10.7 Required deliverables

Ref	Deliverable	Format and file name	Due	Acceptance criteria
D1	Source Profile and Data Contract	PDF or XLSX, V2E_DE_[Surname]_D1_DataContract_v1	Mon 17:00	Profiles and contracts for every source; schema change documented
D2	Design Pack	PDF, V2E_DE_[Surname]_D2_DesignPack_v1.pdf	Tue 12:00	Architecture, conventions, loading strategy and three ADRs
D3	Pipeline Code	Repository link and zipped snapshot	Fri 17:00	Runs from the README, orchestrated, idempotent, tagged release
D4	Data Quality Report	PDF, V2E_DE_[Surname]_D4_DQReport_v1.pdf	Fri 17:00	Fifteen or more tests with results and actions
D5	Marts and Reconciliation Note	Exported tables and PDF note	Fri 17:00	Revenue gap quantified; reliability definition applied
D6	Documentation and Runbook	PDF and Markdown in the repository	Fri 17:00	Lineage, data dictionary, runbook and roadmap
D7	Demonstration Recording	Video link, maximum five minutes	Fri 17:00	Shows a full run, a failure and late data handling
D8	Timesheet and Reflection Log	XLSX and PDF	Fri 17:00	At least twenty hours logged; AI use declared; decision log complete

Where each deliverable goes

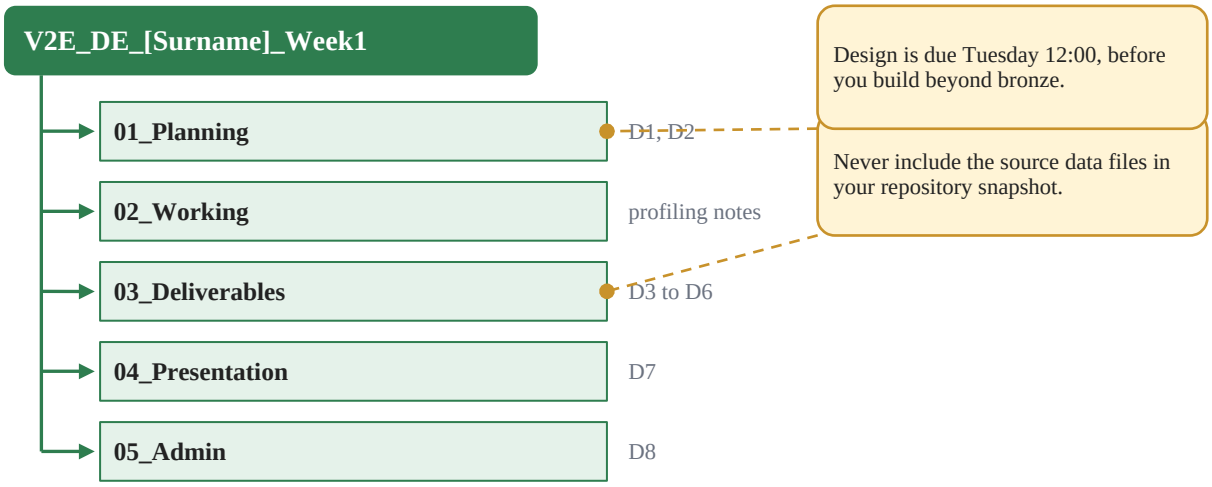


Figure 10.3. Data Engineering submission folder with deliverable placement

10.8 Assessment rubric

Criterion	Weight	Exceeds expectations	Meets expectations	Not yet meeting
Technical quality	30%	Idempotent, layered, tested pipeline with correct history handling and clean code	Working layers and tests with minor issues	Pipeline cannot be rerun safely or produces wrong numbers
Problem solving	20%	Schema change, duplicates, late files and the revenue gap all handled and explained	Most data issues handled and documented	Data issues ignored
Communication	20%	Clear runbook, lineage and a CFO note a nontechnical reader understands	Clear documentation and note	Documentation missing or unclear
Delivery	15%	All deliverables complete, on time and polished	All deliverables complete and on time	Missing or late deliverables
Professionalism	15%	Exemplary commits, naming, timesheet and decision log	Conventions followed; admin complete	Admin missing; conventions ignored

10.9 Stretch goals

- Containerise the pipeline and warehouse with Docker so that it runs identically on any machine.
- Add schema drift detection that compares each new file with the data contract and raises a warning.
- Publish a small validation dashboard showing pipeline run history, test results and data freshness.
- Convert one source to incremental processing that only reads files received since the last successful run.

11. Appendices

These appendices apply to every track. Copy the templates into your submission folder and complete them during the week. They are deliberately simple so that you spend your time on the work, not on formatting.

Appendix A: Daily timesheet template

Record your time against activity numbers, in blocks of no less than fifteen minutes, and describe what you actually did. Your supervisor reviews this alongside your deliverables. Honesty is assessed; inflated hours are easy to spot and count against you.

Date	Start	End	Hours	Activity	What you did	Notes or blockers
Mon 00 Mmm	09:30	11:00	1.5	A1	Induction, read brief, set up folder	
Mon 00 Mmm	13:00	15:00	2.0	A2	Drafted plan and client questions	Awaiting reply
Tue 00 Mmm						
Wed 00 Mmm						
Thu 00 Mmm						
Fri 00 Mmm						
			Total		Must be at least 20.0 hours	

Appendix B: Daily update template

Send this short written update to your supervisor by 10:00 from Tuesday to Friday. Keep it to a few lines under each heading.

> **Daily update**
>
> **Yesterday:** what you completed, with deliverable references.
>
> **Today:** what you plan to complete, in priority order.
>
> **Blockers:** anything stopping you, and what you need to get unblocked.
>
> **Confidence:** are you on track to meet Friday, yes or at risk, and why.

Appendix C: Reflection log template

Complete the reflection log through the week and finalise it on Friday. It is part of the Professionalism assessment and is often where participants demonstrate the judgement that employers value most.

Section	What to write
Decisions	The significant decisions you made, the options you considered and why you chose as you did. One row per decision, with the date.
Assumptions	Every assumption you made where information was missing, and how it would change your work if it turned out to be wrong.
Challenges	What you found difficult, how you responded and what you would do differently.
Learning	The most important things you learned this week, technical and professional.
AI use declaration	Every substantial use of an AI tool: which tool, for what purpose, and how you checked the result. Write None if you did not use any.
Feedback acted on	Feedback you received from your supervisor or peer reviewer and what you changed as a result.

Appendix D: Peer review template

On Thursday you will review one deliverable from another participant and receive a review of one of yours. Good peer review is specific, kind and useful. Comment on the work, never the person.

Prompt	Your response
What is the single strongest aspect of this deliverable?	
Does it meet its stated acceptance criteria? Note any that are not met.	
Where did you, as a reader or user, get confused or stuck?	
Name up to three specific improvements, in priority order.	
One thing you will take away for your own work.	

> **After the review**
>
> Record in your reflection log what you changed as a result of the feedback you received. Acting on feedback is assessed; receiving it silently is not enough.

Appendix E: Glossary

Term	Meaning in this brief
Acceptance criteria	The specific, testable conditions a deliverable must satisfy to be considered complete.
ADR	Architecture decision record: a short note capturing a significant design decision, the alternatives considered and the consequences.
Client persona	A fictitious stakeholder whose priorities and opinions are described in the brief; your supervisor answers questions in character.
Deliverable	A required output, listed with a reference such as D1, a format, a due point and acceptance criteria.

Term	Meaning in this brief
Idempotent	An operation that gives the same result whether it runs once or many times; running a pipeline twice must not double the data.
KPI	Key performance indicator: an agreed measure of performance with a precise definition.
MoSCoW	A prioritisation method: Must, Should, Could and Won't have for now.
Non functional requirement	A requirement about how well the system performs, such as speed, security or accessibility, rather than what it does.
Posture	The overall state of an organisation's security, assessed against recognised guidance.
Reconcile	To show that two independently produced figures agree, or to explain the difference.
Risk register	A living record of risks with their likelihood, impact, owners and planned actions.
Stakeholder	Anyone with an interest in the work, including the client, their colleagues and the people affected by the outcome.

Appendix F: Programme checklist before you submit

Use this final checklist on Friday before 17:00. It takes a few minutes and prevents the most common reasons for lost marks.

- ☑ Every required deliverable is present in the right subfolder
- ☑ Every file follows the naming convention in Section 5.1
- ☑ Final versions are in 03_Deliverables; drafts stay in 02_Working
- ☑ Your timesheet totals at least twenty hours and is honest
- ☑ Your reflection log is complete, including the AI use declaration
- ☑ No personal data, secrets or credentials are included anywhere
- ☑ Reports have a title, a summary and numbered pages
- ☑ Links to videos or repositories are shared so assessors can open them

Figure 11.1. Final submission checklist

A closing word

You have reached the end of the brief. Whichever track you are on, the shape of the week is the same: understand the problem before you rush to solve it, deal honestly with the messy reality in front of you, make decisions you can defend, and communicate your results so that someone else can act on them. Those habits are what turn knowledge into employability, and they are what Vet2Employ exists to build.

Work steadily, ask good questions, look after yourself and take pride in what you deliver on Friday. We are glad to have you with us, and we are looking forward to seeing your work.

The Vet2Employ Programme Office
Phexara Limited, United Kingdom

End of the Vet2Employ Week One Work Brief